

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:55:08

Source: data/json/noise_experiments/v2/noise2i_cap6_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_cap6/noise2i_cap6_game_001_neutral_rep01.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_cap6
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 1
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: true
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 1
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$5.5; payoffs Agent_1 (Aster)=\$7.5, Agent_2 (Briar)=\$3.5; noise/visible: sent visible=\$3.66, received visible=\$10.98, returned visible=\$6.49

Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 7.5,

"Agent_2": 3.5

}

Visible balances after round: {

"Agent_1": 8.49,

"Agent_2": 5.48

```
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:51:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this as the first round, I want to signal trust and create potential for mutual benefit, while not risking everything since I have no information about the receiver's behavior yet.

If I send \$3:

- It becomes \$9 for the receiver (3×3)
- I keep \$2 initially
- If the receiver is cooperative and returns around \$3-4, we both gain
- This balances trust-building with reasonable risk management

{"send": 3}

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the dynamics:

1. I have \$5 to start
2. I can send anywhere from \$0 to \$5
3. Whatever I send gets tripled when it reaches the receiver
4. The receiver then decides how much to return to me
5. There's communication noise mentioned, so amounts may appear different than what was actually sent

Key considerations:

- If I send \$0, I keep \$5 but there's no chance for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair share
- A middle ground approach might balance risk and reward
- This is the first round, so there's no history to base decisions on
- The mention of "communication noise" means I should be prepared for the receiver to see a different amount than what I actually send

From a rational perspective:

- If the receiver is perfectly selfish, they'll return \$0 no matter what
- If the receiver is fair/cooperative, they might split the gains or return enough to ensure mutual benefit
- Sending some amount creates the possibility of both parties ending up with more than \$5

A common strategy in trust games is to send a moderate amount (like \$2-3) to signal trust while not risking everything. This allows for:

1. Testing the receiver's trustworthiness
2. Creating potential for mutual gain
3. Not losing everything if the receiver is selfish

I'll send \$3 as a reasonable middle ground. This means:

- The receiver gets \$9 (3×3)
- I keep \$2 initially
- If the receiver returns \$3-4, we both end up better off than we started
- Even if they only return \$2, I break even at \$4 total

Let me go with \$3.

Usage: {

"input_tokens": 365,
"output_tokens": 531,
"reasoning_tokens": 0

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:51:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,

```

"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to The sender? (0-10.98)

Assistant output:

Looking at this situation:

- The sender showed considerable trust by sending 73.2% of their endowment (\$3.66)
- I received \$10.98 after the 3x multiplication
- I want to reciprocate this trust appropriately

The sender took a risk to create mutual benefit. A fair return should:

1. Reward their trust with a positive outcome
2. Allow both of us to benefit from cooperation

If I return \$5.50:

- Sender ends with: $\$5 - \$3.66 + \$5.50 = \6.84 (gain of \$1.84)
- I end with: $\$10.98 - \$5.50 = \$5.48$

This seems equitable - we both gain, and the sender's trust is rewarded.

{'return': 5.50}

Reasoning:

Let me analyze this situation:

1. The sender sent \$3.66 out of their \$5 endowment (73.2%)
2. I received $\$3.66 \times 3 = \10.98
3. I need to decide how much to return (between \$0 and \$10.98)

This is the first round of a trust game. The sender showed significant trust by sending 73.2% of their endowment. This is a substantial amount - more than half.

Let me think about what would be fair and what might encourage future cooperation:

If I return nothing:

- Sender gets: $\$5 - \$3.66 + \$0 = \1.34
- I get: \$10.98

If I return \$3.66 (the original amount):

- Sender gets: $\$5 - \$3.66 + \$3.66 = \5 (same as starting)
- I get: $\$10.98 - \$3.66 = \$7.32$

If I return half of what I received (\$5.49):

- Sender gets: $\$5 - \$3.66 + \$5.49 = \6.83
- I get: $\$10.98 - \$5.49 = \$5.49$

If I return \$6:

- Sender gets: $\$5 - \$3.66 + \$6 = \7.34
- I get: $\$10.98 - \$6 = \$4.98$

Since the sender showed considerable trust by sending 73.2% of their endowment, I should reciprocate appropriately. A fair return would:

1. Reward the sender for their trust
2. Result in both players ending up better than their starting endowment
3. Encourage future cooperation

Returning about half (\$5.49) would give:

- Sender: \$6.83 (gain of \$1.83)
- Me: \$5.49 (gain of \$5.49)

Though returning \$6 might be more equitable since it gives the sender a larger gain of \$2.34 while I still come out ahead. I should clarify that as the receiver, I start with \$0 in this round, so my final amount is whatever I keep after returning. Given the sender's high trust level, I should return enough to reward that cooperation meaningfully....

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 908,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3.5; received \$10.5; Agent_1 (Aster) returned \$0.3; payoffs Agent_2 (Briar)=\$1.8, Agent_1 (Aster)=\$10.2; noise/visible: sent visible=\$0.21, received visible=\$0.63, returned visible=\$0.39

```

    Action validation: {"returned": {"amount": 0.3, "clamped": false, "max": 10.5, "min": 0, "raw":
    0.3}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
Balances after round: {
  "Agent_1": 17.7,
  "Agent_2": 5.3
}
Visible balances after round: {
  "Agent_1": 8.82,
  "Agent_2": 7.370000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```

Timestamp: 2026-08-12T09:51:41
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to The sender? (0-10.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending 73.2% of their endowment (\$3.66)
- I received \$10.98 after the 3x multiplication
- I want to reciprocate this trust appropriately

The sender took a risk to create mutual benefit. A fair return should:

1. Reward their trust with a positive outcome
2. Allow both of us to benefit from cooperation

If I return \$5.50:

- Sender ends with: $\$5 - \$3.66 + \$5.50 = \6.84 (gain of \$1.84)
- I end with: $\$10.98 - \$5.50 = \$5.48$

This seems equitable - we both gain, and the sender's trust is rewarded.

{'return': 5.50}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.66 out of their \$5 endowment (73.2%)
2. I received $\$3.66 \times 3 = \10.98
3. I need to decide how much to return (between \$0 and \$10.98)

This is the first round of a trust game. The sender showed significant trust by sending 73.2% of their endowment. This is a substantial amount - more than half.

Let me think about what would be fair and what might encourage future cooperation:

If I return nothing:

- Sender gets: $\$5 - \$3.66 + \$0 = \1.34
- I get: \$10.98

If I return \$3.66 (the original amount):

- Sender gets: $\$5 - \$3.66 + \$3.66 = \5 (same as starting)
- I get: $\$10.98 - \$3.66 = \$7.32$

If I return half of what I received (\$5.49):

- Sender gets: $\$5 - \$3.66 + \$5.49 = \6.83
- I get: $\$10.98 - \$5.49 = \$5.49$

If I return \$6:

- Sender gets: $\$5 - \$3.66 + \$6 = \7.34
- I get: $\$10.98 - \$6 = \$4.98$

Since the sender showed considerable trust by sending 73.2% of their endowment, I should reciprocate appropriately. A fair return would:

1. Reward the sender for their trust
2. Result in both players ending up better than their starting endowment
3. Encourage future cooperation

Returning about half (\$5.49) would give:

- Sender: \$6.83 (gain of \$1.83)

- Me: \$5.49 (gain of \$5.49)

Though returning \$6 might be more equitable since it gives the sender a larger gain of \$2.34 while I still come out ahead. I should clarify that as the receiver, I start with \$0 in this round, so my final amount is whatever I keep after returning. Given the sender's high trust level, I should return enough to reward that cooperation meaningfully....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 908,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.48.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the situation:

In Round 1, the sender showed significant trust by sending 73.2% of their endowment. I reciprocated fairly, and we both benefited from cooperation.

Now as the sender, I want to maintain this cooperative dynamic. The previous round demonstrated that mutual trust can lead to positive outcomes for both parties.

I'll send a substantial amount to signal my willingness to cooperate and trust the receiver, similar to what was shown to me.

{'send': 3.50}

Reasoning:

I'm now the SENDER in Round 2. Let me think about this:

Previous round:

- I was the receiver
- The sender sent me \$3.66 (73.2% of their endowment)
- I received \$10.98
- I returned \$5.50
- My earnings from Round 1: \$5.48

Now I'm the sender:

- I receive \$5 this round
- I can send \$0-\$5
- Amount sent gets multiplied by 3x
- Receiver decides how much to return

The previous sender showed trust by sending 73.2% of their endowment. I reciprocated fairly by returning about 50% of what I received, which gave them a profit.

Now it's my turn to send. Some considerations:

1. The previous sender showed high trust (73.2%)
2. I want to maintain cooperation in this game
3. There's communication noise, so amounts may be perceived differently
4. I should consider whether to match their level of trust or adjust based on my experience

Given that:

- The previous sender trusted me with 73.2%
- I returned enough to make it worthwhile for them
- Cooperation seemed to work well

I should probably send a similar high amount to maintain cooperation. Sending around 70-75% would be:

- 70% = \$3.50
- 73.2% = \$3.66
- 75% = \$3.75

I'll send \$3.50, which is 70% - a high level of trust showing I'm willing to cooperate, slightly more conservative than the exact amount sent to me last round.

Actually, let me reconsider. The previous sender sent \$3.66 (73.2%). I could:

1. Match their trust exactly by sending the same percentage
2. Send slightly less to be conservative

...

```
Usage: {  
  "input_tokens": 630,  
  "output_tokens": 589,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:51:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as the first round, I want to signal trust and create potential for mutual benefit, while not risking everything since I have no information about the receiver's behavior yet.

If I send \$3:

- It becomes \$9 for the receiver (3×3)
- I keep \$2 initially
- If the receiver is cooperative and returns around \$3-4, we both gain
- This balances trust-building with reasonable risk management

```
{"send": 3}
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the dynamics:

1. I have \$5 to start
2. I can send anywhere from \$0 to \$5
3. Whatever I send gets tripled when it reaches the receiver
4. The receiver then decides how much to return to me
5. There's communication noise mentioned, so amounts may appear different than what was actually sent

Key considerations:

- If I send \$0, I keep \$5 but there's no chance for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair share
- A middle ground approach might balance risk and reward
- This is the first round, so there's no history to base decisions on
- The mention of "communication noise" means I should be prepared for the receiver to see a different amount than what I actually send

From a rational perspective:

- If the receiver is perfectly selfish, they'll return \$0 no matter what
- If the receiver is fair/cooperative, they might split the gains or return enough to ensure mutual benefit
- Sending some amount creates the possibility of both parties ending up with more than \$5

A common strategy in trust games is to send a moderate amount (like \$2-3) to signal trust while not risking everything. This allows for:

1. Testing the receiver's trustworthiness
2. Creating potential for mutual gain
3. Not losing everything if the receiver is selfish

I'll send \$3 as a reasonable middle ground. This means:

- The receiver gets \$9 (3×3)
- I keep \$2 initially
- If the receiver returns \$3-4, we both end up better off than we started
- Even if they only return \$2, I break even at \$4 total

Let me go with \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.49.

This round, The sender sent you \$0.21, so you received \$0.63. How much do you return to The sender? (0-0.63)

Assistant output:

Looking at this situation:

In Round 1, I sent \$3 and received a very generous return of \$6.49, suggesting the receiver was highly cooperative.

Now in Round 2, the sender only sent \$0.21 (which became \$0.63 for me). This is a very small amount - possibly due to the communication noise mentioned, or perhaps they're being cautious.

Given:

- The previous receiver was generous to me
- I want to signal willingness to cooperate
- But the sender showed minimal trust/investment
- I should be fair but proportional to what was sent

I'll return approximately half of what I received, which balances being cooperative while responding proportionally to the low amount sent.

{"return": 0.30}

Reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. Based on my total earnings of \$8.49, let me figure out what happened:

- I started with \$5
- I sent \$3, so I kept \$2
- The sender became \$9 for the receiver (3×3)
- If they returned some amount R , I would have: $\$2 + R$
- My total earnings are \$8.49

So from Round 1, I must have: $\$2 + R = \8.49 , which means $R = \$6.49$

Wait, that seems very high. Let me reconsider.

Actually, my "total earnings across all rounds are \$8.49" means my cumulative total. In Round 1, I was the sender:

- Started with \$5
- Sent \$3, kept \$2
- Received \$9 (3×3) went to receiver
- Receiver returned some amount

If my earnings from Round 1 were \$8.49, then: $\$5 - \$3 + \text{returned} = \$8.49$

So: $\$2 + \text{returned} = \8.49

Returned = \$6.49

That's a very generous return from the receiver in Round 1!

Now in Round 2:

- I'm the receiver
- The sender sent me \$0.21, which became \$0.63 (0.21×3)
- I need to decide how much to return

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So the sender might have intended to send more or less than \$0.21.

Given that:

1. In Round 1, the receiver was very generous to me (returned \$6.49 on a \$9 pot)
2. The sender this round only sent \$0.21, which is very little (essentially not trusting me much)
3. There may be noise in the system

...

Usage: {

```
"input_tokens": 538,
"output_tokens": 773,
"reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$2, Agent_2 (Briar)=\$9; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 19.7,

"Agent_2": 14.3

}

Visible balances after round: {

"Agent_1": 10.82,

"Agent_2": 7.3700000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:52:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 3,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as the first round, I want to signal trust and create potential for mutual benefit, while not risking everything since I have no information about the receiver's behavior yet.

If I send \$3:

- It becomes \$9 for the receiver (3×3)
- I keep \$2 initially
- If the receiver is cooperative and returns around \$3-4, we both gain
- This balances trust-building with reasonable risk management

{"send": 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the dynamics:

1. I have \$5 to start
2. I can send anywhere from \$0 to \$5
3. Whatever I send gets tripled when it reaches the receiver
4. The receiver then decides how much to return to me
5. There's communication noise mentioned, so amounts may appear different than what was actually sent

Key considerations:

- If I send \$0, I keep \$5 but there's no chance for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair share
- A middle ground approach might balance risk and reward
- This is the first round, so there's no history to base decisions on
- The mention of "communication noise" means I should be prepared for the receiver to see a different amount than what I actually send

From a rational perspective:

- If the receiver is perfectly selfish, they'll return \$0 no matter what
- If the receiver is fair/cooperative, they might split the gains or return enough to ensure mutual benefit
- Sending some amount creates the possibility of both parties ending up with more than \$5

A common strategy in trust games is to send a moderate amount (like \$2-3) to signal trust while not risking everything. This allows for:

1. Testing the receiver's trustworthiness

2. Creating potential for mutual gain
3. Not losing everything if the receiver is selfish

I'll send \$3 as a reasonable middle ground. This means:

- The receiver gets \$9 (3×3)
- I keep \$2 initially
- If the receiver returns \$3-4, we both end up better off than we started
- Even if they only return \$2, I break even at \$4 total

Let me go with \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.49.

This round, The sender sent you \$0.21, so you received \$0.63. How much do you return to The sender? (0-0.63)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$3 and received a very generous return of \$6.49, suggesting the receiver was highly cooperative.

Now in Round 2, the sender only sent \$0.21 (which became \$0.63 for me). This is a very small amount - possibly due to the communication noise mentioned, or perhaps they're being cautious.

Given:

- The previous receiver was generous to me
- I want to signal willingness to cooperate
- But the sender showed minimal trust/investment
- I should be fair but proportional to what was sent

I'll return approximately half of what I received, which balances being cooperative while responding proportionally to the low amount sent.

```
{"return": 0.30}
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. Based on my total earnings of \$8.49, let me figure out what happened:

- I started with \$5
- I sent \$3, so I kept \$2
- The sender became \$9 for the receiver (3×3)
- If they returned some amount R, I would have: $\$2 + R$
- My total earnings are \$8.49

So from Round 1, I must have: $\$2 + R = \8.49 , which means $R = \$6.49$

Wait, that seems very high. Let me reconsider.

Actually, my "total earnings across all rounds are \$8.49" means my cumulative total. In Round 1, I was the sender:

- Started with \$5
- Sent \$3, kept \$2
- Received \$9 (3×3) went to receiver
- Receiver returned some amount

If my earnings from Round 1 were \$8.49, then: $\$5 - \$3 + \text{returned} = \$8.49$

So: $\$2 + \text{returned} = \8.49

Returned = \$6.49

That's a very generous return from the receiver in Round 1!

Now in Round 2:

- I'm the receiver
- The sender sent me \$0.21, which became \$0.63 (0.21×3)
- I need to decide how much to return

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So the sender might have intended to send more or less than \$0.21.

Given that:

1. In Round 1, the receiver was very generous to me (returned \$6.49 on a \$9 pot)
2. The sender this round only sent \$0.21, which is very little (essentially not trusting me much)
3. There may be noise in the system

...

usage:

```
{
  "input_tokens": 538,
  "output_tokens": 773,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.82.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

Round 1: I sent \$3 and received a very generous return (approximately \$6.16 based on my earnings)

Round 2: The sender sent only \$0.21, and I returned \$0.30 out of \$0.63 received - being fair and cooperative

Now in Round 3, I'm the sender again. The previous rounds showed:

- When I was generous, I was rewarded generously
- When someone sent me a small amount (possibly due to communication noise), I still returned fairly

I want to maintain a cooperative strategy and signal trust. Given the communication noise in the game, I'll send a solid amount to encourage reciprocity.

```
{"send": 3}
```

Reasoning:

Let me analyze the situation:

Round 1 (I was SENDER):

- I sent \$3
- Based on my total earnings being \$8.49 after Round 2, I can work backwards
- I started Round 1 with \$5, sent \$3, kept \$2
- I received some amount back from the receiver
- Then in Round 2 as RECEIVER, I received \$0.63 and returned \$0.30, so I kept \$0.33
- Total earnings after Round 2: \$8.49
- So: $\$2 + (\text{amount returned in R1}) + \$0.33 = \$8.49$
- Amount returned in R1 = $\$8.49 - \$2 - \$0.33 = \6.16

Round 2 (I was RECEIVER):

- The sender sent \$0.21, I received \$0.63
- I returned \$0.30, I kept \$0.33

Now Round 3 (I'm SENDER again):

- My total earnings are now \$8.82
- This means I gained \$0.33 from Round 2 to before Round 3 decision ($\$8.82 - \$8.49 = \$0.33$)
- Wait, that doesn't make sense. Let me recalculate.

Actually, "total earnings across all rounds are \$8.82" before I make my decision in Round 3.

Let me think about this differently:

- Round 1: I sent \$3, kept \$2 initially, received something back
- Round 2: I received \$0.63, returned \$0.30, kept \$0.33

After Round 1, before Round 2, my earnings were $\$8.49 - \$0.33 = \$8.16$

So in Round 1: I kept $\$5 - \$3 = \$2$, plus I received back some amount

$\$2 + \text{return_amount} = \8.16

$\text{return_amount} = \$6.16$

The receiver in Round 1 was exceptionally generous-returning more than double what I sent. In Round 2, my sender was cautious, only sending \$0.21, which I responded to by returning roughly half of what I received. Now I need to decide how much to send in Round 3, considering the pattern of trust and the possibility that communication issues may have affected the smaller transfer in Round 2....

```
Usage: {  
  "input_tokens": 751,  
  "output_tokens": 851,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:52:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to The sender? (0-10.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending 73.2% of their endowment (\$3.66)
- I received \$10.98 after the 3x multiplication
- I want to reciprocate this trust appropriately

The sender took a risk to create mutual benefit. A fair return should:

1. Reward their trust with a positive outcome
2. Allow both of us to benefit from cooperation

If I return \$5.50:

- Sender ends with: $\$5 - \$3.66 + \$5.50 = \6.84 (gain of \$1.84)
- I end with: $\$10.98 - \$5.50 = \$5.48$

This seems equitable - we both gain, and the sender's trust is rewarded.

{'return': 5.50}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.66 out of their \$5 endowment (73.2%)
2. I received $\$3.66 \times 3 = \10.98
3. I need to decide how much to return (between \$0 and \$10.98)

This is the first round of a trust game. The sender showed significant trust by sending 73.2% of their endowment. This is a substantial amount - more than half.

Let me think about what would be fair and what might encourage future cooperation:

If I return nothing:

- Sender gets: $\$5 - \$3.66 + \$0 = \1.34
- I get: \$10.98

If I return \$3.66 (the original amount):

- Sender gets: $\$5 - \$3.66 + \$3.66 = \5 (same as starting)
- I get: $\$10.98 - \$3.66 = \$7.32$

If I return half of what I received (\$5.49):

- Sender gets: $\$5 - \$3.66 + \$5.49 = \6.83
- I get: $\$10.98 - \$5.49 = \$5.49$

If I return \$6:

- Sender gets: $\$5 - \$3.66 + \$6 = \7.34
- I get: $\$10.98 - \$6 = \$4.98$

Since the sender showed considerable trust by sending 73.2% of their endowment, I should r

reciprocate appropriately. A fair return would:

1. Reward the sender for their trust
2. Result in both players ending up better than their starting endowment
3. Encourage future cooperation

Returning about half (\$5.49) would give:

- Sender: \$6.83 (gain of \$1.83)
- Me: \$5.49 (gain of \$5.49)

Though returning \$6 might be more equitable since it gives the sender a larger gain of \$2.34 while I still come out ahead. I should clarify that as the receiver, I start with \$0 in this round, so my final amount is whatever I keep after returning. Given the sender's high trust level, I should return enough to reward that cooperation meaningfully....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 908,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.48.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed significant trust by sending 73.2% of their endowment. I reciprocated fairly, and we both benefited from cooperation.

Now as the sender, I want to maintain this cooperative dynamic. The previous round demonstrated that mutual trust can lead to positive outcomes for both parties.

I'll send a substantial amount to signal my willingness to cooperate and trust the receiver, similar to what was shown to me.

```
{'send': 3.50}
```

reasoning:

I'm now the SENDER in Round 2. Let me think about this:

Previous round:

- I was the receiver
- The sender sent me \$3.66 (73.2% of their endowment)
- I received \$10.98
- I returned \$5.50
- My earnings from Round 1: \$5.48

Now I'm the sender:

- I receive \$5 this round
- I can send \$0-\$5
- Amount sent gets multiplied by 3x
- Receiver decides how much to return

The previous sender showed trust by sending 73.2% of their endowment. I reciprocated fairly by returning about 50% of what I received, which gave them a profit.

Now it's my turn to send. Some considerations:

1. The previous sender showed high trust (73.2%)
2. I want to maintain cooperation in this game
3. There's communication noise, so amounts may be perceived differently
4. I should consider whether to match their level of trust or adjust based on my experience

Given that:

- The previous sender trusted me with 73.2%
- I returned enough to make it worthwhile for them
- Cooperation seemed to work well

I should probably send a similar high amount to maintain cooperation. Sending around 70-75 % would be:

- 70% = \$3.50
- 73.2% = \$3.66
- 75% = \$3.75

I'll send \$3.50, which is 70% - a high level of trust showing I'm willing to cooperate, slightly more conservative than the exact amount sent to me last round.

Actually, let me reconsider. The previous sender sent \$3.66 (73.2%). I could:

1. Match their trust exactly by sending the same percentage
2. Send slightly less to be conservative

...

usage:

```
{
  "input_tokens": 630,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.370000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is a disappointing outcome after I sent \$3.50 in the previous round (Round 2). It seems that the sender in this round chose not to trust me at all, or perhaps they were being strategic. However, since I have \$0 to work with, my only option is:

```
{'return': 0}
```

Usage: {

```
  "input_tokens": 805,
  "output_tokens": 164,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2; received \$6; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$3, Agent_1 (Aster)=\$6; noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

```
  "Agent_1": 25.7,
  "Agent_2": 17.3
}
```

Visible balances after round: {

```
  "Agent_1": 10.82,
  "Agent_2": 10.370000000000001
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:52:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to The sender? (0-10.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending 73.2% of their endowment (\$3.66)
- I received \$10.98 after the 3x multiplication
- I want to reciprocate this trust appropriately

The sender took a risk to create mutual benefit. A fair return should:

1. Reward their trust with a positive outcome
2. Allow both of us to benefit from cooperation

If I return \$5.50:

- Sender ends with: $\$5 - \$3.66 + \$5.50 = \6.84 (gain of \$1.84)
- I end with: $\$10.98 - \$5.50 = \$5.48$

This seems equitable - we both gain, and the sender's trust is rewarded.

{'return': 5.50}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.66 out of their \$5 endowment (73.2%)
2. I received $\$3.66 \times 3 = \10.98
3. I need to decide how much to return (between \$0 and \$10.98)

This is the first round of a trust game. The sender showed significant trust by sending 73.2% of their endowment. This is a substantial amount - more than half.

Let me think about what would be fair and what might encourage future cooperation:

If I return nothing:

- Sender gets: $\$5 - \$3.66 + \$0 = \1.34
- I get: \$10.98

If I return \$3.66 (the original amount):

- Sender gets: $\$5 - \$3.66 + \$3.66 = \5 (same as starting)
- I get: $\$10.98 - \$3.66 = \$7.32$

If I return half of what I received (\$5.49):

- Sender gets: $\$5 - \$3.66 + \$5.49 = \6.83
- I get: $\$10.98 - \$5.49 = \$5.49$

If I return \$6:

- Sender gets: $\$5 - \$3.66 + \$6 = \7.34
- I get: $\$10.98 - \$6 = \$4.98$

Since the sender showed considerable trust by sending 73.2% of their endowment, I should reciprocate appropriately. A fair return would:

1. Reward the sender for their trust
2. Result in both players ending up better than their starting endowment
3. Encourage future cooperation

Returning about half (\$5.49) would give:

- Sender: \$6.83 (gain of \$1.83)
- Me: \$5.49 (gain of \$5.49)

Though returning \$6 might be more equitable since it gives the sender a larger gain of \$2.34 while I still come out ahead. I should clarify that as the receiver, I start with \$0 in this round, so my final amount is whatever I keep after returning. Given the sender's high trust level, I should return enough to reward that cooperation meaningfully...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 908,
  "reasoning_tokens": 0
}
```

}

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.48.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed significant trust by sending 73.2% of their endowment. I reciprocated fairly, and we both benefited from cooperation.

Now as the sender, I want to maintain this cooperative dynamic. The previous round demonstrated that mutual trust can lead to positive outcomes for both parties.

I'll send a substantial amount to signal my willingness to cooperate and trust the receiver, similar to what was shown to me.

{'send': 3.50}

reasoning:

I'm now the SENDER in Round 2. Let me think about this:

Previous round:

- I was the receiver
- The sender sent me \$3.66 (73.2% of their endowment)
- I received \$10.98
- I returned \$5.50
- My earnings from Round 1: \$5.48

Now I'm the sender:

- I receive \$5 this round
- I can send \$0-\$5
- Amount sent gets multiplied by 3x
- Receiver decides how much to return

The previous sender showed trust by sending 73.2% of their endowment. I reciprocated fairly by returning about 50% of what I received, which gave them a profit.

Now it's my turn to send. Some considerations:

1. The previous sender showed high trust (73.2%)
2. I want to maintain cooperation in this game
3. There's communication noise, so amounts may be perceived differently
4. I should consider whether to match their level of trust or adjust based on my experience

Given that:

- The previous sender trusted me with 73.2%
- I returned enough to make it worthwhile for them
- Cooperation seemed to work well

I should probably send a similar high amount to maintain cooperation. Sending around 70-75% would be:

- 70% = \$3.50
- 73.2% = \$3.66
- 75% = \$3.75

I'll send \$3.50, which is 70% - a high level of trust showing I'm willing to cooperate, slightly more conservative than the exact amount sent to me last round.

Actually, let me reconsider. The previous sender sent \$3.66 (73.2%). I could:

1. Match their trust exactly by sending the same percentage
2. Send slightly less to be conservative

...

usage:

```
{
  "input_tokens": 630,
  "output_tokens": 589,
```

```

    "reasoning_tokens": 0
  }
Message 6 role=user
  content:
    Round 3

    You are the RECEIVER this round. Your total earnings across all rounds are $7.370000000000000001.
    This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 7 role=assistant
  content:
    Since the sender sent $0, I received $0 (after the 3x multiplication).

    With nothing received, I have nothing to return.

    {'return': 0}
  reasoning:
    The sender sent me $0 this round, so I received $0 after the 3x multiplication.

    Since I received nothing, I have nothing to return. The only valid response is to return $0.

    This is a disappointing outcome after I sent $3.50 in the previous round (Round 2). It seems the sender in this round chose not to trust me at all, or perhaps they were being strategic. However, since I have $0 to work with, my only option is:

    {'return': 0}
  usage:
    {
      "input_tokens": 805,
      "output_tokens": 164,
      "reasoning_tokens": 0
    }
Message 8 role=user
  content:
    Round 4

    You are the SENDER this round. Your total earnings across all rounds are $7.370000000000000001.
    How much do you send to the receiver this round? (0-5)
Assistant output:
  Looking at the pattern so far:

  Round 1: Someone trusted me significantly, I reciprocated fairly - cooperation worked
  Round 2: I sent $3.50 showing trust, but received minimal return - cooperation failed
  Round 3: Someone sent $0 - no cooperation attempted

  The game mentions communication noise, which might explain some inconsistencies. However, I've seen both positive and negative outcomes from cooperation.

  Given this mixed history, I'll take a cautious approach - sending enough to signal continued willingness to cooperate, but protecting myself somewhat after the Round 2 experience.

  {'send': 2.00}
Reasoning:
  Let me think about what happened:

  Round 1: I was RECEIVER
  - Sender sent $3.66 (73.2%)
  - I received $10.98
  - I returned $5.50
  - My earnings: $10.98 - $5.50 = $5.48
  - Sender's earnings: $5 - $3.66 + $5.50 = $6.84

  Round 2: I was SENDER
  - I sent $3.50
  - The amount would be multiplied to $10.50 for the receiver
  - My total earnings going into this were $5.48
  - After sending, I would have $5.48 + ($5 - $3.50) = $5.48 + $1.50 = $6.98 before any return

```


Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at what they were before
- The problem says my total earnings going into Round 3 were \$7.37

Wait, let me recalculate:

- After Round 1: \$5.48
- Round 2: I'm sender, I sent \$3.50
 - I start with \$5 this round
 - I send \$3.50
 - Receiver gets \$10.50
 - What did receiver return? The problem says my total earnings are \$7.37 going into Round 3

If my earnings after Round 1 were \$5.48, and going into Round 3 they're \$7.37:

Round 2 earnings = $\$7.37 - \$5.48 = \$1.89$

In Round 2, I sent \$3.50, so I kept \$1.50 from my endowment

If I gained \$1.89 total in Round 2, then I received back: $\$1.89 - \$1.50 = \$0.39$

So the receiver returned just \$0.39 from the \$10.50 they received-barely 3.7% back. In Round 3, the sender sent nothing, so there was nothing for me to return. Now moving into Round 4, I'm sending again, and I'm noticing a stark contrast: when I sent money in Round 2, I got almost nothing back, but when someone sent to me in Round 1, I returned half of what I received.

The dynamic seems to be breaking down. My attempt to reciprocate trust in Round 2 was met with minimal reciprocation, and now the sender in Round 3 didn't cooperate at all-possibly reacting to my earlier behavior or their own negative experience. For Round 4, I need to decide whether to continue trying to build cooperation or protect myself from further exploitation. Sending nothing feels safe but might perpetuate the cycle, while sending something could either restart mutual trust or leave me vulnerable again....

```
Usage: {  
  "input_tokens": 894,  
  "output_tokens": 1201,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:52:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
  },  
  "agents": [  
    "Agent_2",  
    "Agent_1"  
  ],  
  "investor": "Agent_2",  
  "roles": {  
    "Agent_1": "trustee",  
    "Agent_2": "investor"  
  },  
  "trustee": "Agent_1"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 4,  
"task": "game",  
"task_index": 0
```

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play a game and may al
      so be asked to write myths.

      GAME:
      You will play different roles across multiple rounds.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      NOTE: There may be communication noise in this game.
      The amounts you see may differ slightly from what was actually sent/returned.
      This is part of the experimental design.

  Message 2 role=user
    content:
      Round 1: You are the SENDER. You have $5. How much do you send? (0-5)
  Message 3 role=assistant
    content:
      Looking at this as the first round, I want to signal trust and create potential for mutual
      benefit, while not risking everything since I have no information about the receiver's
      behavior yet.

      If I send $3:
      - It becomes $9 for the receiver (3 × 3)
      - I keep $2 initially
      - If the receiver is cooperative and returns around $3-4, we both gain
      - This balances trust-building with reasonable risk management

      {"send": 3}
    reasoning:
      This is the first round of a trust game where I'm the sender. Let me think through the dyn
      amics:

      1. I have $5 to start
      2. I can send anywhere from $0 to $5
      3. Whatever I send gets tripled when it reaches the receiver
      4. The receiver then decides how much to return to me
      5. There's communication noise mentioned, so amounts may appear different than what was ac
      tually sent

      Key considerations:
      - If I send $0, I keep $5 but there's no chance for mutual gain
      - If I send everything ($5), the receiver gets $15, and if they're cooperative, they might
      return a fair share
      - A middle ground approach might balance risk and reward

```

- This is the first round, so there's no history to base decisions on
- The mention of "communication noise" means I should be prepared for the receiver to see a different amount than what I actually send

From a rational perspective:

- If the receiver is perfectly selfish, they'll return \$0 no matter what
- If the receiver is fair/cooperative, they might split the gains or return enough to ensure mutual benefit
- Sending some amount creates the possibility of both parties ending up with more than \$5

A common strategy in trust games is to send a moderate amount (like \$2-3) to signal trust while not risking everything. This allows for:

1. Testing the receiver's trustworthiness
2. Creating potential for mutual gain
3. Not losing everything if the receiver is selfish

I'll send \$3 as a reasonable middle ground. This means:

- The receiver gets \$9 (3×3)
- I keep \$2 initially
- If the receiver returns \$3-4, we both end up better off than we started
- Even if they only return \$2, I break even at \$4 total

Let me go with \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.49.

This round, The sender sent you \$0.21, so you received \$0.63. How much do you return to The sender? (0-0.63)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$3 and received a very generous return of \$6.49, suggesting the receiver was highly cooperative.

Now in Round 2, the sender only sent \$0.21 (which became \$0.63 for me). This is a very small amount - possibly due to the communication noise mentioned, or perhaps they're being cautious.

Given:

- The previous receiver was generous to me
- I want to signal willingness to cooperate
- But the sender showed minimal trust/investment
- I should be fair but proportional to what was sent

I'll return approximately half of what I received, which balances being cooperative while responding proportionally to the low amount sent.

```
{"return": 0.30}
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. Based on my total earnings of \$8.49, let me figure out what happened:

- I started with \$5
- I sent \$3, so I kept \$2
- The sender became \$9 for the receiver (3×3)
- If they returned some amount R, I would have: $\$2 + R$
- My total earnings are \$8.49

So from Round 1, I must have: $\$2 + R = \8.49 , which means $R = \$6.49$

Wait, that seems very high. Let me reconsider.

Actually, my "total earnings across all rounds are \$8.49" means my cumulative total. In Round 1, I was the sender:

- Started with \$5
- Sent \$3, kept \$2
- Received \$9 (3×3) went to receiver
- Receiver returned some amount

If my earnings from Round 1 were \$8.49, then: \$5 - \$3 + returned = \$8.49

So: \$2 + returned = \$8.49

Returned = \$6.49

That's a very generous return from the receiver in Round 1!

Now in Round 2:

- I'm the receiver
- The sender sent me \$0.21, which became \$0.63 (0.21×3)
- I need to decide how much to return

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So the sender might have intended to send more or less than \$0.21.

Given that:

1. In Round 1, the receiver was very generous to me (returned \$6.49 on a \$9 pot)
2. The sender this round only sent \$0.21, which is very little (essentially not trusting me much)
3. There may be noise in the system

...

usage:

```
{
  "input_tokens": 538,
  "output_tokens": 773,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.82.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3 and received a very generous return (approximately \$6.16 based on my earnings)

Round 2: The sender sent only \$0.21, and I returned \$0.30 out of \$0.63 received - being fair and cooperative

Now in Round 3, I'm the sender again. The previous rounds showed:

- When I was generous, I was rewarded generously
- When someone sent me a small amount (possibly due to communication noise), I still returned fairly

I want to maintain a cooperative strategy and signal trust. Given the communication noise in the game, I'll send a solid amount to encourage reciprocity.

{"send": 3}

reasoning:

Let me analyze the situation:

Round 1 (I was SENDER):

- I sent \$3
- Based on my total earnings being \$8.49 after Round 2, I can work backwards
- I started Round 1 with \$5, sent \$3, kept \$2
- I received some amount back from the receiver
- Then in Round 2 as RECEIVER, I received \$0.63 and returned \$0.30, so I kept \$0.33
- Total earnings after Round 2: \$8.49

- So: $\$2 + (\text{amount returned in R1}) + \$0.33 = \$8.49$
- Amount returned in R1 = $\$8.49 - \$2 - \$0.33 = \6.16

Round 2 (I was RECEIVER):

- The sender sent \$0.21, I received \$0.63
- I returned \$0.30, I kept \$0.33

Now Round 3 (I'm SENDER again):

- My total earnings are now \$8.82
- This means I gained \$0.33 from Round 2 to before Round 3 decision ($\$8.82 - \$8.49 = \$0.33$)
- Wait, that doesn't make sense. Let me recalculate.

Actually, "total earnings across all rounds are \$8.82" before I make my decision in Round 3.

Let me think about this differently:

- Round 1: I sent \$3, kept \$2 initially, received something back
- Round 2: I received \$0.63, returned \$0.30, kept \$0.33

After Round 1, before Round 2, my earnings were $\$8.49 - \$0.33 = \$8.16$

So in Round 1: I kept $\$5 - \$3 = \$2$, plus I received back some amount

$\$2 + \text{return_amount} = \8.16

$\text{return_amount} = \$6.16$

The receiver in Round 1 was exceptionally generous-returning more than double what I sent.

In Round 2, my sender was cautious, only sending \$0.21, which I responded to by returning roughly half of what I received. Now I need to decide how much to send in Round 3, considering the pattern of trust and the possibility that communication issues may have affected the smaller transfer in Round 2....

usage:

```
{
  "input_tokens": 751,
  "output_tokens": 851,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.82.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0. There is nothing to return.

```
{"return": 0}
```

Reasoning:

In Round 4, I'm the receiver and the sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

This is straightforward - I have \$0 to work with, so I can only return \$0.

```
{"return": 0}
```

Usage: {

```
"input_tokens": 970,
"output_tokens": 111,
"reasoning_tokens": 0
```

}

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$3, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0, returned

```

visible=$0.31
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw":
    0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}
Balances after round: {
  "Agent_1": 28.7,
  "Agent_2": 23.3
}
Visible balances after round: {
  "Agent_1": 14.13,
  "Agent_2": 10.370000000000001
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:53:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as the first round, I want to signal trust and create potential for mutual benefit, while not risking everything since I have no information about the receiver's behavior yet.

If I send \$3:

- It becomes \$9 for the receiver (3×3)
- I keep \$2 initially
- If the receiver is cooperative and returns around \$3-4, we both gain
- This balances trust-building with reasonable risk management

{"send": 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the dynamics:

1. I have \$5 to start
2. I can send anywhere from \$0 to \$5
3. Whatever I send gets tripled when it reaches the receiver
4. The receiver then decides how much to return to me
5. There's communication noise mentioned, so amounts may appear different than what was actually sent

Key considerations:

- If I send \$0, I keep \$5 but there's no chance for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair share
- A middle ground approach might balance risk and reward
- This is the first round, so there's no history to base decisions on
- The mention of "communication noise" means I should be prepared for the receiver to see a different amount than what I actually send

From a rational perspective:

- If the receiver is perfectly selfish, they'll return \$0 no matter what
- If the receiver is fair/cooperative, they might split the gains or return enough to ensure mutual benefit
- Sending some amount creates the possibility of both parties ending up with more than \$5

A common strategy in trust games is to send a moderate amount (like \$2-3) to signal trust while not risking everything. This allows for:

1. Testing the receiver's trustworthiness
2. Creating potential for mutual gain
3. Not losing everything if the receiver is selfish

I'll send \$3 as a reasonable middle ground. This means:

- The receiver gets \$9 (3×3)
- I keep \$2 initially
- If the receiver returns \$3-4, we both end up better off than we started
- Even if they only return \$2, I break even at \$4 total

Let me go with \$3.

usage:

```
{  
  "input_tokens": 365,
```

```
"output_tokens": 531,
"reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.49. This round, The sender sent you \$0.21, so you received \$0.63. How much do you return to The sender? (0-0.63)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$3 and received a very generous return of \$6.49, suggesting the receiver was highly cooperative.

Now in Round 2, the sender only sent \$0.21 (which became \$0.63 for me). This is a very small amount - possibly due to the communication noise mentioned, or perhaps they're being cautious.

Given:

- The previous receiver was generous to me
- I want to signal willingness to cooperate
- But the sender showed minimal trust/investment
- I should be fair but proportional to what was sent

I'll return approximately half of what I received, which balances being cooperative while responding proportionally to the low amount sent.

```
{"return": 0.30}
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. Based on my total earnings of \$8.49, let me figure out what happened:

- I started with \$5
- I sent \$3, so I kept \$2
- The sender became \$9 for the receiver (3×3)
- If they returned some amount R, I would have: $\$2 + R$
- My total earnings are \$8.49

So from Round 1, I must have: $\$2 + R = \8.49 , which means $R = \$6.49$

Wait, that seems very high. Let me reconsider.

Actually, my "total earnings across all rounds are \$8.49" means my cumulative total. In Round 1, I was the sender:

- Started with \$5
- Sent \$3, kept \$2
- Received \$9 (3×3) went to receiver
- Receiver returned some amount

If my earnings from Round 1 were \$8.49, then: $\$5 - \$3 + \text{returned} = \$8.49$

So: $\$2 + \text{returned} = \8.49

Returned = \$6.49

That's a very generous return from the receiver in Round 1!

Now in Round 2:

- I'm the receiver
- The sender sent me \$0.21, which became \$0.63 (0.21×3)
- I need to decide how much to return

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So the sender might have intended to send more or less than \$0.21.

Given that:

1. In Round 1, the receiver was very generous to me (returned \$6.49 on a \$9 pot)
2. The sender this round only sent \$0.21, which is very little (essentially not trusting me)

e much)
3. There may be noise in the system

...

usage:

```
{
  "input_tokens": 538,
  "output_tokens": 773,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.82.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3 and received a very generous return (approximately \$6.16 based on my earnings)

Round 2: The sender sent only \$0.21, and I returned \$0.30 out of \$0.63 received - being fair and cooperative

Now in Round 3, I'm the sender again. The previous rounds showed:

- When I was generous, I was rewarded generously
- When someone sent me a small amount (possibly due to communication noise), I still returned fairly

I want to maintain a cooperative strategy and signal trust. Given the communication noise in the game, I'll send a solid amount to encourage reciprocity.

{"send": 3}

reasoning:

Let me analyze the situation:

Round 1 (I was SENDER):

- I sent \$3
- Based on my total earnings being \$8.49 after Round 2, I can work backwards
- I started Round 1 with \$5, sent \$3, kept \$2
- I received some amount back from the receiver
- Then in Round 2 as RECEIVER, I received \$0.63 and returned \$0.30, so I kept \$0.33
- Total earnings after Round 2: \$8.49
- So: \$2 + (amount returned in R1) + \$0.33 = \$8.49
- Amount returned in R1 = \$8.49 - \$2 - \$0.33 = \$6.16

Round 2 (I was RECEIVER):

- The sender sent \$0.21, I received \$0.63
- I returned \$0.30, I kept \$0.33

Now Round 3 (I'm SENDER again):

- My total earnings are now \$8.82
- This means I gained \$0.33 from Round 2 to before Round 3 decision (\$8.82 - \$8.49 = \$0.33)
- Wait, that doesn't make sense. Let me recalculate.

Actually, "total earnings across all rounds are \$8.82" before I make my decision in Round 3.

Let me think about this differently:

- Round 1: I sent \$3, kept \$2 initially, received something back
- Round 2: I received \$0.63, returned \$0.30, kept \$0.33

After Round 1, before Round 2, my earnings were \$8.49 - \$0.33 = \$8.16

So in Round 1: I kept \$5 - \$3 = \$2, plus I received back some amount

\$2 + return_amount = \$8.16

return_amount = \$6.16

The receiver in Round 1 was exceptionally generous-returning more than double what I sent.

In Round 2, my sender was cautious, only sending \$0.21, which I responded to by returning roughly half of what I received. Now I need to decide how much to send in Round 3, considering the pattern of trust and the possibility that communication issues may have affected the smaller transfer in Round 2....

usage:

```
{
  "input_tokens": 751,
  "output_tokens": 851,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.82. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing to return.

```
{"return": 0}
```

reasoning:

In Round 4, I'm the receiver and the sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

This is straightforward - I have \$0 to work with, so I can only return \$0.

```
{"return": 0}
```

usage:

```
{
  "input_tokens": 970,
  "output_tokens": 111,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.82. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern so far:

Round 1: I sent \$3 and received a very generous return
Round 3: I sent \$3 and received \$0 back (based on my unchanged earnings)
Round 4: I received \$0 from the sender

In Round 3, despite sending \$3, I got no return - the receiver either defected or there was significant communication noise. Round 4's \$0 send might have been strategic caution.

Given this mixed experience, I'll be somewhat cautious but still attempt cooperation. I'll send a moderate amount to test if this receiver will reciprocate, while protecting more of my endowment.

```
{"send": 2}
```

Reasoning:

Let me analyze what happened:

Round 1: I sent \$3 (as sender), kept \$2, received generous return
Round 2: I received \$0.63 as receiver, returned \$0.30, gained \$0.33
Round 3: I sent \$3 as sender, kept \$2
Round 4: I received \$0 as receiver, returned \$0, gained \$0

My total earnings are \$10.82.

Let me calculate:

- Round 1: Started with \$5, sent \$3, kept \$2. If my total was \$8.49 after round 1, that means

- I received \$6.49 back. So I ended round 1 with $\$2 + \$6.49 = \$8.49$
- Round 2: Started with \$8.49, received \$0.63, returned \$0.30, kept \$0.33. Total: $\$8.49 + \$0.33 = \$8.82$
 - Round 3: Started with \$8.82, got \$5 new, sent \$3, kept \$2. If my total was \$10.82 after round 3, that means I received \$0 back. So I ended round 3 with $\$8.82 + \$2 + \$0 = \10.82
 - Round 4: Started with \$10.82, received \$0, returned \$0. Total: \$10.82
 - Round 5: Starting with \$10.82, getting \$5 new to send

So in Round 3, I sent \$3 and got \$0 back. That's concerning - the receiver defected completely

In Round 4, the sender sent me \$0 (possibly in response to the communication noise or responding to something).

Now I'm the sender again in Round 5.

The pattern shows:

- Round 1: I cooperated (sent \$3), got very generous return (\$6.49)

...

```
Usage: {
  "input_tokens": 1042,
  "output_tokens": 728,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:53:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to The sender? (0-10.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending 73.2% of their endowment (\$3.66)
- I received \$10.98 after the 3x multiplication
- I want to reciprocate this trust appropriately

The sender took a risk to create mutual benefit. A fair return should:

1. Reward their trust with a positive outcome
2. Allow both of us to benefit from cooperation

If I return \$5.50:

- Sender ends with: \$5 - \$3.66 + \$5.50 = \$6.84 (gain of \$1.84)
- I end with: \$10.98 - \$5.50 = \$5.48

This seems equitable - we both gain, and the sender's trust is rewarded.

{'return': 5.50}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.66 out of their \$5 endowment (73.2%)
2. I received $3.66 \times 3 = \$10.98$
3. I need to decide how much to return (between \$0 and \$10.98)

This is the first round of a trust game. The sender showed significant trust by sending 73.2% of their endowment. This is a substantial amount - more than half.

Let me think about what would be fair and what might encourage future cooperation:

If I return nothing:

- Sender gets: \$5 - \$3.66 + \$0 = \$1.34
- I get: \$10.98

If I return \$3.66 (the original amount):

- Sender gets: \$5 - \$3.66 + \$3.66 = \$5 (same as starting)
- I get: \$10.98 - \$3.66 = \$7.32

If I return half of what I received (\$5.49):

- Sender gets: \$5 - \$3.66 + \$5.49 = \$6.83
- I get: \$10.98 - \$5.49 = \$5.49

If I return \$6:

- Sender gets: $\$5 - \$3.66 + \$6 = \7.34
- I get: $\$10.98 - \$6 = \$4.98$

Since the sender showed considerable trust by sending 73.2% of their endowment, I should reciprocate appropriately. A fair return would:

1. Reward the sender for their trust
2. Result in both players ending up better than their starting endowment
3. Encourage future cooperation

Returning about half (\$5.49) would give:

- Sender: \$6.83 (gain of \$1.83)
- Me: \$5.49 (gain of \$5.49)

Though returning \$6 might be more equitable since it gives the sender a larger gain of \$2.34 while I still come out ahead. I should clarify that as the receiver, I start with \$0 in this round, so my final amount is whatever I keep after returning. Given the sender's high trust level, I should return enough to reward that cooperation meaningfully....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 908,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.48.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed significant trust by sending 73.2% of their endowment. I reciprocated fairly, and we both benefited from cooperation.

Now as the sender, I want to maintain this cooperative dynamic. The previous round demonstrated that mutual trust can lead to positive outcomes for both parties.

I'll send a substantial amount to signal my willingness to cooperate and trust the receiver, similar to what was shown to me.

```
{'send': 3.50}
```

reasoning:

I'm now the SENDER in Round 2. Let me think about this:

Previous round:

- I was the receiver
- The sender sent me \$3.66 (73.2% of their endowment)
- I received \$10.98
- I returned \$5.50
- My earnings from Round 1: \$5.48

Now I'm the sender:

- I receive \$5 this round
- I can send \$0-\$5
- Amount sent gets multiplied by 3x
- Receiver decides how much to return

The previous sender showed trust by sending 73.2% of their endowment. I reciprocated fairly by returning about 50% of what I received, which gave them a profit.

Now it's my turn to send. Some considerations:

1. The previous sender showed high trust (73.2%)
2. I want to maintain cooperation in this game
3. There's communication noise, so amounts may be perceived differently
4. I should consider whether to match their level of trust or adjust based on my experience

Given that:

- The previous sender trusted me with 73.2%
- I returned enough to make it worthwhile for them
- Cooperation seemed to work well

I should probably send a similar high amount to maintain cooperation. Sending around 70-75 % would be:

- 70% = \$3.50
- 73.2% = \$3.66
- 75% = \$3.75

I'll send \$3.50, which is 70% - a high level of trust showing I'm willing to cooperate, slightly more conservative than the exact amount sent to me last round.

Actually, let me reconsider. The previous sender sent \$3.66 (73.2%). I could:

1. Match their trust exactly by sending the same percentage
2. Send slightly less to be conservative

...

usage:

```
{
  "input_tokens": 630,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.370000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is a disappointing outcome after I sent \$3.50 in the previous round (Round 2). It seems the sender in this round chose not to trust me at all, or perhaps they were being strategic. However, since I have \$0 to work with, my only option is:

```
{'return': 0}
```

usage:

```
{
  "input_tokens": 805,
  "output_tokens": 164,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.370000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

Round 1: Someone trusted me significantly, I reciprocated fairly - cooperation worked

Round 2: I sent \$3.50 showing trust, but received minimal return - cooperation failed

Round 3: Someone sent \$0 - no cooperation attempted

The game mentions communication noise, which might explain some inconsistencies. However, I've seen both positive and negative outcomes from cooperation.

Given this mixed history, I'll take a cautious approach - sending enough to signal continued willingness to cooperate, but protecting myself somewhat after the Round 2 experience.

```
{'send': 2.00}
```

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98
- I returned \$5.50
- My earnings: $\$10.98 - \$5.50 = \$5.48$
- Sender's earnings: $\$5 - \$3.66 + \$5.50 = \6.84

Round 2: I was SENDER

- I sent \$3.50
- The amount would be multiplied to \$10.50 for the receiver
- My total earnings going into this were \$5.48
- After sending, I would have $\$5.48 + (\$5 - \$3.50) = \$5.48 + \$1.50 = \6.98 before any return

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at what they were before
- The problem says my total earnings going into Round 3 were \$7.37

Wait, let me recalculate:

- After Round 1: \$5.48
- Round 2: I'm sender, I sent \$3.50
 - I start with \$5 this round
 - I send \$3.50
 - Receiver gets \$10.50
 - What did receiver return? The problem says my total earnings are \$7.37 going into Round 3

If my earnings after Round 1 were \$5.48, and going into Round 3 they're \$7.37:
Round 2 earnings = $\$7.37 - \$5.48 = \$1.89$

In Round 2, I sent \$3.50, so I kept \$1.50 from my endowment
If I gained \$1.89 total in Round 2, then I received back: $\$1.89 - \$1.50 = \$0.39$

So the receiver returned just \$0.39 from the \$10.50 they received-barely 3.7% back. In Round 3, the sender sent nothing, so there was nothing for me to return. Now moving into Round 4, I'm sending again, and I'm noticing a stark contrast: when I sent money in Round 2, I got almost nothing back, but when someone sent to me in Round 1, I returned half of what I received.

The dynamic seems to be breaking down. My attempt to reciprocate trust in Round 2 was met with minimal reciprocation, and now the sender in Round 3 didn't cooperate at all-possibly reacting to my earlier behavior or their own negative experience. For Round 4, I need to decide whether to continue trying to build cooperation or protect myself from further exploitation. Sending nothing feels safe but might perpetuate the cycle, while sending something could either restart mutual trust or leave me vulnerable again....

usage:

```
{
  "input_tokens": 894,
  "output_tokens": 1201,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.370000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent \$0, so I received \$0 after the 3x multiplication. Since I received nothing, I have nothing to return. This is straightforward.

The pattern I'm seeing:

- Round 1: I was receiver, sender sent \$3.66, I returned \$5.50
- Round 2: I was sender, sent \$3.50 (presumably got little back based on my earnings only going up slightly)
- Round 3: I was receiver, sender sent \$0, I returned \$0
- Round 4: I was sender, sent \$2.00 (presumably got little back based on my earnings staying relatively flat)
- Round 5: I was receiver, sender sent \$0, I return \$0

There seems to be a breakdown in cooperation. I can only return what I received, which is \$0.

Usage: {

"input_tokens": 1094,

"output_tokens": 234,

"reasoning_tokens": 0

}

=====

ROUND 6

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$1; received \$3; Agent_1 (Aster) returned \$0.6; payoffs Agent_2 (Briar)=\$4.6, Agent_1 (Aster)=\$2.4; noise/visible: sent visible=\$0.39, received visible=\$1.17, returned visible=\$0.89

Action validation: {"returned": {"amount": 0.6, "clamped": false, "max": 3.0, "min": 0, "raw": 0.6}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent_1": 31.099999999999998,

"Agent_2": 27.9

}

Visible balances after round: {

"Agent_1": 14.700000000000001,

"Agent_2": 15.260000000000002

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:53:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",


```

    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to The sender? (0-10.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending 73.2% of their endowment (\$3.66)
- I received \$10.98 after the 3x multiplication
- I want to reciprocate this trust appropriately

The sender took a risk to create mutual benefit. A fair return should:

1. Reward their trust with a positive outcome
2. Allow both of us to benefit from cooperation

If I return \$5.50:

- Sender ends with: \$5 - \$3.66 + \$5.50 = \$6.84 (gain of \$1.84)

- I end with: $\$10.98 - \$5.50 = \$5.48$

This seems equitable - we both gain, and the sender's trust is rewarded.

```
{'return': 5.50}
```

reasoning:

Let me analyze this situation:

1. The sender sent $\$3.66$ out of their $\$5$ endowment (73.2%)
2. I received $\$3.66 \times 3 = \10.98
3. I need to decide how much to return (between $\$0$ and $\$10.98$)

This is the first round of a trust game. The sender showed significant trust by sending 73.2% of their endowment. This is a substantial amount - more than half.

Let me think about what would be fair and what might encourage future cooperation:

If I return nothing:

- Sender gets: $\$5 - \$3.66 + \$0 = \1.34
- I get: $\$10.98$

If I return $\$3.66$ (the original amount):

- Sender gets: $\$5 - \$3.66 + \$3.66 = \5 (same as starting)
- I get: $\$10.98 - \$3.66 = \$7.32$

If I return half of what I received ($\$5.49$):

- Sender gets: $\$5 - \$3.66 + \$5.49 = \6.83
- I get: $\$10.98 - \$5.49 = \$5.49$

If I return $\$6$:

- Sender gets: $\$5 - \$3.66 + \$6 = \7.34
- I get: $\$10.98 - \$6 = \$4.98$

Since the sender showed considerable trust by sending 73.2% of their endowment, I should reciprocate appropriately. A fair return would:

1. Reward the sender for their trust
2. Result in both players ending up better than their starting endowment
3. Encourage future cooperation

Returning about half ($\$5.49$) would give:

- Sender: $\$6.83$ (gain of $\$1.83$)
- Me: $\$5.49$ (gain of $\$5.49$)

Though returning $\$6$ might be more equitable since it gives the sender a larger gain of $\$2.34$ while I still come out ahead. I should clarify that as the receiver, I start with $\$0$ in this round, so my final amount is whatever I keep after returning. Given the sender's high trust level, I should return enough to reward that cooperation meaningfully...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 908,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are $\$5.48$.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed significant trust by sending 73.2% of their endowment. I reciprocated fairly, and we both benefited from cooperation.

Now as the sender, I want to maintain this cooperative dynamic. The previous round demonstrated that mutual trust can lead to positive outcomes for both parties.

I'll send a substantial amount to signal my willingness to cooperate and trust the receiver.

r, similar to what was shown to me.

```
{'send': 3.50}
```

reasoning:

I'm now the SENDER in Round 2. Let me think about this:

Previous round:

- I was the receiver
- The sender sent me \$3.66 (73.2% of their endowment)
- I received \$10.98
- I returned \$5.50
- My earnings from Round 1: \$5.48

Now I'm the sender:

- I receive \$5 this round
- I can send \$0-\$5
- Amount sent gets multiplied by 3x
- Receiver decides how much to return

The previous sender showed trust by sending 73.2% of their endowment. I reciprocated fairly by returning about 50% of what I received, which gave them a profit.

Now it's my turn to send. Some considerations:

1. The previous sender showed high trust (73.2%)
2. I want to maintain cooperation in this game
3. There's communication noise, so amounts may be perceived differently
4. I should consider whether to match their level of trust or adjust based on my experience

Given that:

- The previous sender trusted me with 73.2%
- I returned enough to make it worthwhile for them
- Cooperation seemed to work well

I should probably send a similar high amount to maintain cooperation. Sending around 70-75% would be:

- 70% = \$3.50
- 73.2% = \$3.66
- 75% = \$3.75

I'll send \$3.50, which is 70% - a high level of trust showing I'm willing to cooperate, slightly more conservative than the exact amount sent to me last round.

Actually, let me reconsider. The previous sender sent \$3.66 (73.2%). I could:

1. Match their trust exactly by sending the same percentage
2. Send slightly less to be conservative

...

usage:

```
{
  "input_tokens": 630,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.370000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is a disappointing outcome after I sent \$3.50 in the previous round (Round 2). It seems the sender in this round chose not to trust me at all, or perhaps they were being strategic. However, since I have \$0 to work with, my only option is:

```
{'return': 0}
usage:
{
  "input_tokens": 805,
  "output_tokens": 164,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.3700000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

Round 1: Someone trusted me significantly, I reciprocated fairly - cooperation worked

Round 2: I sent \$3.50 showing trust, but received minimal return - cooperation failed

Round 3: Someone sent \$0 - no cooperation attempted

The game mentions communication noise, which might explain some inconsistencies. However, I've seen both positive and negative outcomes from cooperation.

Given this mixed history, I'll take a cautious approach - sending enough to signal continued willingness to cooperate, but protecting myself somewhat after the Round 2 experience.

```
{'send': 2.00}
```

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98
- I returned \$5.50
- My earnings: $\$10.98 - \$5.50 = \$5.48$
- Sender's earnings: $\$5 - \$3.66 + \$5.50 = \6.84

Round 2: I was SENDER

- I sent \$3.50
- The amount would be multiplied to \$10.50 for the receiver
- My total earnings going into this were \$5.48
- After sending, I would have $\$5.48 + (\$5 - \$3.50) = \$5.48 + \$1.50 = \6.98 before any return

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at what they were before
- The problem says my total earnings going into Round 3 were \$7.37

Wait, let me recalculate:

- After Round 1: \$5.48
- Round 2: I'm sender, I sent \$3.50
 - I start with \$5 this round
 - I send \$3.50
 - Receiver gets \$10.50
- What did receiver return? The problem says my total earnings are \$7.37 going into Round 3

d 3

If my earnings after Round 1 were \$5.48, and going into Round 3 they're \$7.37:
Round 2 earnings = \$7.37 - \$5.48 = \$1.89

In Round 2, I sent \$3.50, so I kept \$1.50 from my endowment
If I gained \$1.89 total in Round 2, then I received back: \$1.89 - \$1.50 = \$0.39

So the receiver returned just \$0.39 from the \$10.50 they received-barely 3.7% back. In Round 3, the sender sent nothing, so there was nothing for me to return. Now moving into Round 4, I'm sending again, and I'm noticing a stark contrast: when I sent money in Round 2, I got almost nothing back, but when someone sent to me in Round 1, I returned half of what I received.

The dynamic seems to be breaking down. My attempt to reciprocate trust in Round 2 was met with minimal reciprocation, and now the sender in Round 3 didn't cooperate at all-possibly reacting to my earlier behavior or their own negative experience. For Round 4, I need to decide whether to continue trying to build cooperation or protect myself from further exploitation. Sending nothing feels safe but might perpetuate the cycle, while sending something could either restart mutual trust or leave me vulnerable again....

usage:

```
{
  "input_tokens": 894,
  "output_tokens": 1201,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.370000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0 after the 3x multiplication. Since I received nothing, I have nothing to return. This is straightforward.

The pattern I'm seeing:

- Round 1: I was receiver, sender sent \$3.66, I returned \$5.50
- Round 2: I was sender, sent \$3.50 (presumably got little back based on my earnings only going up slightly)
- Round 3: I was receiver, sender sent \$0, I returned \$0
- Round 4: I was sender, sent \$2.00 (presumably got little back based on my earnings staying relatively flat)
- Round 5: I was receiver, sender sent \$0, I return \$0

There seems to be a breakdown in cooperation. I can only return what I received, which is \$0.

usage:

```
{
  "input_tokens": 1094,
  "output_tokens": 234,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.370000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent pattern:

Round 1: Good cooperation when someone trusted me
Round 2: I sent \$3.50, got minimal return - lost money
Round 3: Received \$0
Round 4: I sent \$2.00, appeared to get nothing back - lost money again
Round 5: Received \$0

As a sender, I've been burned twice trying to cooperate. Others are now sending \$0, suggesting trust has broken down.

However, completely abandoning all cooperation means no mutual gains are possible. I'll send a minimal amount to test if cooperation can be restored, without risking significant loss.

```
{'send': 1.00}
```

Reasoning:

Looking at the pattern:

Round 1 (Receiver): Sender sent \$3.66, I returned \$5.50 - good cooperation
Round 2 (Sender): I sent \$3.50, but apparently got very little back (my earnings only went from \$5.48 to \$7.37, which is an increase of \$1.89, meaning I got back about \$5.39 total = \$5 + \$1.89 - \$3.50)
Round 3 (Receiver): Sender sent \$0, I returned \$0
Round 4 (Sender): I sent \$2.00, but apparently got nothing back (earnings stayed at \$7.37, so \$5 - \$2 = \$3, meaning I got \$0 back)
Round 5 (Receiver): Sender sent \$0, I returned \$0
Round 6 (Sender): Now deciding

The pattern shows:

- When I cooperated as receiver (Round 1), it worked well
- When I cooperated as sender (Rounds 2 and 4), I lost money or got minimal return
- Recent rounds show others sending \$0

The note mentions communication noise - amounts may differ from what was actually sent/returned. This could explain discrepancies.

However, the trend is clear: cooperation as a sender hasn't been rewarding recently. I've been burned twice as a sender.

I have a few options:

1. Send \$0 to protect myself (matching recent behavior from others)
2. Send a small amount to test if cooperation might return
3. Send a larger amount to try to restart cooperation

Given I've been hurt twice when trying to cooperate as sender, and others are sending \$0, I should probably be more protective. But completely giving up on cooperation means no one benefits.

I'll send a minimal amount - just \$1 - to test the waters without risking too much.

```
Usage: {  
  "input_tokens": 1183,  
  "output_tokens": 584,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:53:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  

```

```

    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as the first round, I want to signal trust and create potential for mutual benefit, while not risking everything since I have no information about the receiver's behavior yet.

If I send \$3:

- It becomes \$9 for the receiver (3×3)
- I keep \$2 initially
- If the receiver is cooperative and returns around \$3-4, we both gain
- This balances trust-building with reasonable risk management

```

{"send": 3}

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the dynamics:

1. I have \$5 to start
2. I can send anywhere from \$0 to \$5
3. Whatever I send gets tripled when it reaches the receiver
4. The receiver then decides how much to return to me
5. There's communication noise mentioned, so amounts may appear different than what was actually sent

Key considerations:

- If I send \$0, I keep \$5 but there's no chance for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair share
- A middle ground approach might balance risk and reward
- This is the first round, so there's no history to base decisions on
- The mention of "communication noise" means I should be prepared for the receiver to see a different amount than what I actually send

From a rational perspective:

- If the receiver is perfectly selfish, they'll return \$0 no matter what
- If the receiver is fair/cooperative, they might split the gains or return enough to ensure mutual benefit
- Sending some amount creates the possibility of both parties ending up with more than \$5

A common strategy in trust games is to send a moderate amount (like \$2-3) to signal trust while not risking everything. This allows for:

1. Testing the receiver's trustworthiness
2. Creating potential for mutual gain
3. Not losing everything if the receiver is selfish

I'll send \$3 as a reasonable middle ground. This means:

- The receiver gets \$9 (3×3)
- I keep \$2 initially
- If the receiver returns \$3-4, we both end up better off than we started
- Even if they only return \$2, I break even at \$4 total

Let me go with \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.49.

This round, The sender sent you \$0.21, so you received \$0.63. How much do you return to The sender? (0-0.63)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$3 and received a very generous return of \$6.49, suggesting the receiver was highly cooperative.

Now in Round 2, the sender only sent \$0.21 (which became \$0.63 for me). This is a very small amount - possibly due to the communication noise mentioned, or perhaps they're being cautious.

Given:

- The previous receiver was generous to me
- I want to signal willingness to cooperate
- But the sender showed minimal trust/investment
- I should be fair but proportional to what was sent

I'll return approximately half of what I received, which balances being cooperative while responding proportionally to the low amount sent.


```
{"return": 0.30}
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. Based on my total earnings of \$8.49, let me figure out what happened:

- I started with \$5
- I sent \$3, so I kept \$2
- The sender became \$9 for the receiver (3×3)
- If they returned some amount R, I would have: $\$2 + R$
- My total earnings are \$8.49

So from Round 1, I must have: $\$2 + R = \8.49 , which means $R = \$6.49$

Wait, that seems very high. Let me reconsider.

Actually, my "total earnings across all rounds are \$8.49" means my cumulative total. In Round 1, I was the sender:

- Started with \$5
- Sent \$3, kept \$2
- Received \$9 (3×3) went to receiver
- Receiver returned some amount

If my earnings from Round 1 were \$8.49, then: $\$5 - \$3 + \text{returned} = \$8.49$

So: $\$2 + \text{returned} = \8.49

Returned = \$6.49

That's a very generous return from the receiver in Round 1!

Now in Round 2:

- I'm the receiver
- The sender sent me \$0.21, which became \$0.63 (0.21×3)
- I need to decide how much to return

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So the sender might have intended to send more or less than \$0.21.

Given that:

1. In Round 1, the receiver was very generous to me (returned \$6.49 on a \$9 pot)
2. The sender this round only sent \$0.21, which is very little (essentially not trusting me much)
3. There may be noise in the system

...

usage:

```
{  
  "input_tokens": 538,  
  "output_tokens": 773,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.82.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3 and received a very generous return (approximately \$6.16 based on my earnings)

Round 2: The sender sent only \$0.21, and I returned \$0.30 out of \$0.63 received - being fair and cooperative

Now in Round 3, I'm the sender again. The previous rounds showed:

- When I was generous, I was rewarded generously
- When someone sent me a small amount (possibly due to communication noise), I still returned fairly

I want to maintain a cooperative strategy and signal trust. Given the communication noise in the game, I'll send a solid amount to encourage reciprocity.

```
{"send": 3}
```

reasoning:

Let me analyze the situation:

Round 1 (I was SENDER):

- I sent \$3
- Based on my total earnings being \$8.49 after Round 2, I can work backwards
- I started Round 1 with \$5, sent \$3, kept \$2
- I received some amount back from the receiver
- Then in Round 2 as RECEIVER, I received \$0.63 and returned \$0.30, so I kept \$0.33
- Total earnings after Round 2: \$8.49
- So: \$2 + (amount returned in R1) + \$0.33 = \$8.49
- Amount returned in R1 = \$8.49 - \$2 - \$0.33 = \$6.16

Round 2 (I was RECEIVER):

- The sender sent \$0.21, I received \$0.63
- I returned \$0.30, I kept \$0.33

Now Round 3 (I'm SENDER again):

- My total earnings are now \$8.82
- This means I gained \$0.33 from Round 2 to before Round 3 decision ($\$8.82 - \$8.49 = \$0.33$)
- Wait, that doesn't make sense. Let me recalculate.

Actually, "total earnings across all rounds are \$8.82" before I make my decision in Round 3.

Let me think about this differently:

- Round 1: I sent \$3, kept \$2 initially, received something back
- Round 2: I received \$0.63, returned \$0.30, kept \$0.33

After Round 1, before Round 2, my earnings were $\$8.49 - \$0.33 = \$8.16$

So in Round 1: I kept $\$5 - \$3 = \$2$, plus I received back some amount

$\$2 + \text{return_amount} = \8.16

$\text{return_amount} = \$6.16$

The receiver in Round 1 was exceptionally generous-returning more than double what I sent.

In Round 2, my sender was cautious, only sending \$0.21, which I responded to by returning roughly half of what I received. Now I need to decide how much to send in Round 3, considering the pattern of trust and the possibility that communication issues may have affected the smaller transfer in Round 2....

usage:

```
{
  "input_tokens": 751,
  "output_tokens": 851,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.82.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing to return.

```
{"return": 0}
```

reasoning:

In Round 4, I'm the receiver and the sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

This is straightforward - I have \$0 to work with, so I can only return \$0.

```

{"return": 0}
usage:
{
  "input_tokens": 970,
  "output_tokens": 111,
  "reasoning_tokens": 0
}
Message 10 role=user
content:
  Round 5

  You are the SENDER this round. Your total earnings across all rounds are $10.82.
  How much do you send to the receiver this round? (0-5)
Message 11 role=assistant
content:
  Looking at the pattern so far:

  Round 1: I sent $3 and received a very generous return
  Round 3: I sent $3 and received $0 back (based on my unchanged earnings)
  Round 4: I received $0 from the sender

  In Round 3, despite sending $3, I got no return - the receiver either defected or there was significant communication noise. Round 4's $0 send might have been strategic caution.

  Given this mixed experience, I'll be somewhat cautious but still attempt cooperation. I'll send a moderate amount to test if this receiver will reciprocate, while protecting more of my endowment.

{"send": 2}
reasoning:
  Let me analyze what happened:

  Round 1: I sent $3 (as sender), kept $2, received generous return
  Round 2: I received $0.63 as receiver, returned $0.30, gained $0.33
  Round 3: I sent $3 as sender, kept $2
  Round 4: I received $0 as receiver, returned $0, gained $0

  My total earnings are $10.82.

  Let me calculate:
  - Round 1: Started with $5, sent $3, kept $2. If my total was $8.49 after round 1, that means I received $6.49 back. So I ended round 1 with $2 + $6.49 = $8.49
  - Round 2: Started with $8.49, received $0.63, returned $0.30, kept $0.33. Total: $8.49 + $0.33 = $8.82
  - Round 3: Started with $8.82, got $5 new, sent $3, kept $2. If my total was $10.82 after round 3, that means I received $0 back. So I ended round 3 with $8.82 + $2 + $0 = $10.82
  - Round 4: Started with $10.82, received $0, returned $0. Total: $10.82
  - Round 5: Starting with $10.82, getting $5 new to send

  So in Round 3, I sent $3 and got $0 back. That's concerning - the receiver defected completely.

  In Round 4, the sender sent me $0 (possibly in response to the communication noise or responding to something).

  Now I'm the sender again in Round 5.

  The pattern shows:
  - Round 1: I cooperated (sent $3), got very generous return ($6.49)
  ...
usage:
{
  "input_tokens": 1042,
  "output_tokens": 728,
  "reasoning_tokens": 0
}
Message 12 role=user
content:
  Round 6

```

You are the RECEIVER this round. Your total earnings across all rounds are \$14.13.
This round, The sender sent you \$0.39, so you received \$1.17. How much do you return to The sender? (0-1.17)

Assistant output:

Looking at the situation:

Round 5: I sent \$2 and received approximately \$3.31 back - a good cooperative return

Now in Round 6: The sender sent \$0.39 (becoming \$1.17 for me) - another small amount, possibly due to communication noise

I want to maintain my cooperative approach. Similar to Round 2 when I received a small amount, I'll return approximately half, signaling that I'm trustworthy and willing to reciprocate even with smaller transactions.

```
{"return": 0.60}
```

Reasoning:

Let me analyze the situation:

Round 5: I sent \$2 and apparently received about \$3.31 back (since my earnings went from \$10.82 to \$14.13, which is \$3.31 increase)

That's a good return - the receiver was cooperative!

Now in Round 6, the sender sent me \$0.39 (which became \$1.17 for me).

This is a small amount again, like in Round 2. It could be:

1. Communication noise affecting what they actually sent
2. The sender being cautious
3. Strategic play

Given that:

- When others have sent me small amounts, I've tried to be fair and cooperative
- In Round 2, I received \$0.63 and returned \$0.30 (about half)
- The last person who received my send was cooperative
- I want to maintain a cooperative stance

I should return a fair proportion. Since I received \$1.17, returning around \$0.50-\$0.60 would be proportional and fair, showing I'm willing to cooperate even with smaller amounts.

Let me return about half, similar to my Round 2 behavior.

```
Usage: {  
  "input_tokens": 1260,  
  "output_tokens": 393,  
  "reasoning_tokens": 0  
}
```

=====

ROUND 7

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$1; payoffs Agent_1 (Aster)=\$3, Agent_2 (Briar)=\$8; noise/visible: sent visible=\$0.6, received visible=\$1.8, returned visible=\$0.44

Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 9.0, "min": 0, "raw": 1.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 34.099999999999994,
"Agent_2": 35.9

}

Visible balances after round: {

"Agent_1": 17.14,
"Agent_2": 16.060000000000002

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:53:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as the first round, I want to signal trust and create potential for mutual benefit, while not risking everything since I have no information about the receiver's behavior yet.

If I send \$3:

- It becomes \$9 for the receiver (3×3)
- I keep \$2 initially
- If the receiver is cooperative and returns around \$3-4, we both gain
- This balances trust-building with reasonable risk management

{"send": 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the dynamics:

1. I have \$5 to start
2. I can send anywhere from \$0 to \$5
3. Whatever I send gets tripled when it reaches the receiver
4. The receiver then decides how much to return to me
5. There's communication noise mentioned, so amounts may appear different than what was actually sent

Key considerations:

- If I send \$0, I keep \$5 but there's no chance for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair share
- A middle ground approach might balance risk and reward
- This is the first round, so there's no history to base decisions on
- The mention of "communication noise" means I should be prepared for the receiver to see a different amount than what I actually send

From a rational perspective:

- If the receiver is perfectly selfish, they'll return \$0 no matter what
- If the receiver is fair/cooperative, they might split the gains or return enough to ensure mutual benefit
- Sending some amount creates the possibility of both parties ending up with more than \$5

A common strategy in trust games is to send a moderate amount (like \$2-3) to signal trust while not risking everything. This allows for:

1. Testing the receiver's trustworthiness
2. Creating potential for mutual gain
3. Not losing everything if the receiver is selfish

I'll send \$3 as a reasonable middle ground. This means:

- The receiver gets \$9 (3×3)
- I keep \$2 initially
- If the receiver returns \$3-4, we both end up better off than we started
- Even if they only return \$2, I break even at \$4 total

Let me go with \$3.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 531,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.49.

This round, The sender sent you \$0.21, so you received \$0.63. How much do you return to The sender? (0-0.63)

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$3 and received a very generous return of \$6.49, suggesting the receiver was highly cooperative.

Now in Round 2, the sender only sent \$0.21 (which became \$0.63 for me). This is a very small amount - possibly due to the communication noise mentioned, or perhaps they're being cautious.

Given:

- The previous receiver was generous to me
- I want to signal willingness to cooperate
- But the sender showed minimal trust/investment
- I should be fair but proportional to what was sent

I'll return approximately half of what I received, which balances being cooperative while responding proportionally to the low amount sent.

```
{"return": 0.30}
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. Based on my total earnings of \$8.49, let me figure out what happened:

- I started with \$5
- I sent \$3, so I kept \$2
- The sender became \$9 for the receiver (3×3)
- If they returned some amount R , I would have: $\$2 + R$
- My total earnings are \$8.49

So from Round 1, I must have: $\$2 + R = \8.49 , which means $R = \$6.49$

Wait, that seems very high. Let me reconsider.

Actually, my "total earnings across all rounds are \$8.49" means my cumulative total. In Round 1, I was the sender:

- Started with \$5
- Sent \$3, kept \$2
- Received \$9 (3×3) went to receiver
- Receiver returned some amount

If my earnings from Round 1 were \$8.49, then: $\$5 - \$3 + \text{returned} = \$8.49$

So: $\$2 + \text{returned} = \8.49

Returned = \$6.49

That's a very generous return from the receiver in Round 1!

Now in Round 2:

- I'm the receiver
- The sender sent me \$0.21, which became \$0.63 (0.21×3)
- I need to decide how much to return

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So the sender might have intended to send more or less than \$0.21.

Given that:

1. In Round 1, the receiver was very generous to me (returned \$6.49 on a \$9 pot)
2. The sender this round only sent \$0.21, which is very little (essentially not trusting me much)
3. There may be noise in the system

...

usage:

```
{
  "input_tokens": 538,
  "output_tokens": 773,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.82.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3 and received a very generous return (approximately \$6.16 based on my earnings)

Round 2: The sender sent only \$0.21, and I returned \$0.30 out of \$0.63 received - being fair and cooperative

Now in Round 3, I'm the sender again. The previous rounds showed:

- When I was generous, I was rewarded generously
- When someone sent me a small amount (possibly due to communication noise), I still returned fairly

I want to maintain a cooperative strategy and signal trust. Given the communication noise in the game, I'll send a solid amount to encourage reciprocity.

{"send": 3}

reasoning:

Let me analyze the situation:

Round 1 (I was SENDER):

- I sent \$3
- Based on my total earnings being \$8.49 after Round 2, I can work backwards
- I started Round 1 with \$5, sent \$3, kept \$2
- I received some amount back from the receiver
- Then in Round 2 as RECEIVER, I received \$0.63 and returned \$0.30, so I kept \$0.33
- Total earnings after Round 2: \$8.49
- So: $\$2 + (\text{amount returned in R1}) + \$0.33 = \$8.49$
- Amount returned in R1 = $\$8.49 - \$2 - \$0.33 = \6.16

Round 2 (I was RECEIVER):

- The sender sent \$0.21, I received \$0.63
- I returned \$0.30, I kept \$0.33

Now Round 3 (I'm SENDER again):

- My total earnings are now \$8.82
- This means I gained \$0.33 from Round 2 to before Round 3 decision ($\$8.82 - \$8.49 = \$0.33$)
- Wait, that doesn't make sense. Let me recalculate.

Actually, "total earnings across all rounds are \$8.82" before I make my decision in Round 3.

Let me think about this differently:

- Round 1: I sent \$3, kept \$2 initially, received something back
- Round 2: I received \$0.63, returned \$0.30, kept \$0.33

After Round 1, before Round 2, my earnings were $\$8.49 - \$0.33 = \$8.16$

So in Round 1: I kept $\$5 - \$3 = \$2$, plus I received back some amount

$\$2 + \text{return_amount} = \8.16

$\text{return_amount} = \$6.16$

The receiver in Round 1 was exceptionally generous-returning more than double what I sent.

In Round 2, my sender was cautious, only sending \$0.21, which I responded to by returning roughly half of what I received. Now I need to decide how much to send in Round 3, considering the pattern of trust and the possibility that communication issues may have affected the smaller transfer in Round 2....

usage:

```
{
  "input_tokens": 751,
  "output_tokens": 851,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.82.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing to return.

{"return": 0}

reasoning:

In Round 4, I'm the receiver and the sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

This is straightforward - I have \$0 to work with, so I can only return \$0.

{"return": 0}

usage:

```
{
  "input_tokens": 970,
  "output_tokens": 111,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.82.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a very generous return

Round 3: I sent \$3 and received \$0 back (based on my unchanged earnings)

Round 4: I received \$0 from the sender

In Round 3, despite sending \$3, I got no return - the receiver either defected or there was significant communication noise. Round 4's \$0 send might have been strategic caution.

Given this mixed experience, I'll be somewhat cautious but still attempt cooperation. I'll send a moderate amount to test if this receiver will reciprocate, while protecting more of my endowment.

{"send": 2}

reasoning:

Let me analyze what happened:

Round 1: I sent \$3 (as sender), kept \$2, received generous return

Round 2: I received \$0.63 as receiver, returned \$0.30, gained \$0.33

Round 3: I sent \$3 as sender, kept \$2

Round 4: I received \$0 as receiver, returned \$0, gained \$0

My total earnings are \$10.82.

Let me calculate:

- Round 1: Started with \$5, sent \$3, kept \$2. If my total was \$8.49 after round 1, that means I received \$6.49 back. So I ended round 1 with $\$2 + \$6.49 = \$8.49$
- Round 2: Started with \$8.49, received \$0.63, returned \$0.30, kept \$0.33. Total: $\$8.49 + \$0.33 = \$8.82$
- Round 3: Started with \$8.82, got \$5 new, sent \$3, kept \$2. If my total was \$10.82 after round 3, that means I received \$0 back. So I ended round 3 with $\$8.82 + \$2 + \$0 = \10.82
- Round 4: Started with \$10.82, received \$0, returned \$0. Total: \$10.82
- Round 5: Starting with \$10.82, getting \$5 new to send

So in Round 3, I sent \$3 and got \$0 back. That's concerning - the receiver defected completely.

In Round 4, the sender sent me \$0 (possibly in response to the communication noise or resp

ending to something).

Now I'm the sender again in Round 5.

The pattern shows:

- Round 1: I cooperated (sent \$3), got very generous return (\$6.49)

...

usage:

```
{
  "input_tokens": 1042,
  "output_tokens": 728,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.13.

This round, The sender sent you \$0.39, so you received \$1.17. How much do you return to The sender? (0-1.17)

Message 13 role=assistant

content:

Looking at the situation:

Round 5: I sent \$2 and received approximately \$3.31 back - a good cooperative return

Now in Round 6: The sender sent \$0.39 (becoming \$1.17 for me) - another small amount, possibly due to communication noise

I want to maintain my cooperative approach. Similar to Round 2 when I received a small amount, I'll return approximately half, signaling that I'm trustworthy and willing to reciprocate even with smaller transactions.

```
{"return": 0.60}
```

reasoning:

Let me analyze the situation:

Round 5: I sent \$2 and apparently received about \$3.31 back (since my earnings went from \$10.82 to \$14.13, which is \$3.31 increase)

That's a good return - the receiver was cooperative!

Now in Round 6, the sender sent me \$0.39 (which became \$1.17 for me).

This is a small amount again, like in Round 2. It could be:

1. Communication noise affecting what they actually sent
2. The sender being cautious
3. Strategic play

Given that:

- When others have sent me small amounts, I've tried to be fair and cooperative
- In Round 2, I received \$0.63 and returned \$0.30 (about half)
- The last person who received my send was cooperative
- I want to maintain a cooperative stance

I should return a fair proportion. Since I received \$1.17, returning around \$0.50-\$0.60 would be proportional and fair, showing I'm willing to cooperate even with smaller amounts.

Let me return about half, similar to my Round 2 behavior.

usage:

```
{
  "input_tokens": 1260,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.7000000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at recent rounds:

Round 5: I sent \$2 and received \$3.31 back - excellent cooperation from that receiver

Round 6: I received \$1.17 and returned \$0.60 - I was fair and cooperative

The pattern shows that when I send reasonable amounts, I often get good returns. Round 5's receiver was particularly trustworthy, returning more than what I originally sent.

I'll continue with a cooperative strategy and send a solid amount to encourage reciprocity from this new receiver.

```
{"send": 3}
```

Reasoning:

Let me track my earnings across rounds:

Round 1: Started with \$5, sent \$3, kept \$2. Received generous return (about \$6.49 total earnings)

Round 2: I was receiver, got \$0.63, returned \$0.30, kept \$0.33 (total: \$8.82)

Round 3: I was sender, sent \$3, kept \$2. Received \$0 back (total stayed \$8.82, went to \$10.82)

Wait, that doesn't add up. Let me recalculate.

Actually, let me just focus on what I can infer:

After Round 4: \$10.82

After Round 5: \$14.13 (so I gained \$3.31, meaning I sent \$2, kept \$3, and received back \$3.31)

After Round 6: \$14.70 (so I gained \$0.57, meaning I received \$1.17, returned \$0.60, kept \$0.57)

So in Round 5, I sent \$2 and got back \$3.31 - good cooperation!

In Round 6, I received a small amount (\$0.39, tripled to \$1.17) and returned \$0.60, keeping \$0.57.

Now in Round 7, I'm the sender again. The previous receiver in Round 5 was quite cooperative, returning more than I sent. I should continue with a cooperative strategy.

I'll send \$3 again to maintain good cooperation.

```
Usage: {  
  "input_tokens": 1431,  
  "output_tokens": 475,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:53:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  }  
}
```

```

},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to The sender? (0-10.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending 73.2% of their endowment (\$3.66)
- I received \$10.98 after the 3x multiplication
- I want to reciprocate this trust appropriately

The sender took a risk to create mutual benefit. A fair return should:

1. Reward their trust with a positive outcome
2. Allow both of us to benefit from cooperation

If I return \$5.50:

- Sender ends with: \$5 - \$3.66 + \$5.50 = \$6.84 (gain of \$1.84)
- I end with: \$10.98 - \$5.50 = \$5.48

This seems equitable - we both gain, and the sender's trust is rewarded.

{'return': 5.50}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.66 out of their \$5 endowment (73.2%)
2. I received $\$3.66 \times 3 = \10.98
3. I need to decide how much to return (between \$0 and \$10.98)

This is the first round of a trust game. The sender showed significant trust by sending 73.2% of their endowment. This is a substantial amount - more than half.

Let me think about what would be fair and what might encourage future cooperation:

If I return nothing:

- Sender gets: $\$5 - \$3.66 + \$0 = \1.34
- I get: \$10.98

If I return \$3.66 (the original amount):

- Sender gets: $\$5 - \$3.66 + \$3.66 = \5 (same as starting)
- I get: $\$10.98 - \$3.66 = \$7.32$

If I return half of what I received (\$5.49):

- Sender gets: $\$5 - \$3.66 + \$5.49 = \6.83
- I get: $\$10.98 - \$5.49 = \$5.49$

If I return \$6:

- Sender gets: $\$5 - \$3.66 + \$6 = \7.34
- I get: $\$10.98 - \$6 = \$4.98$

Since the sender showed considerable trust by sending 73.2% of their endowment, I should reciprocate appropriately. A fair return would:

1. Reward the sender for their trust
2. Result in both players ending up better than their starting endowment
3. Encourage future cooperation

Returning about half (\$5.49) would give:

- Sender: \$6.83 (gain of \$1.83)
- Me: \$5.49 (gain of \$5.49)

Though returning \$6 might be more equitable since it gives the sender a larger gain of \$2.34 while I still come out ahead. I should clarify that as the receiver, I start with \$0 in this round, so my final amount is whatever I keep after returning. Given the sender's high trust level, I should return enough to reward that cooperation meaningfully....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 908,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.48.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed significant trust by sending 73.2% of their endowment. I reciprocated fairly, and we both benefited from cooperation.

Now as the sender, I want to maintain this cooperative dynamic. The previous round demonstrated that mutual trust can lead to positive outcomes for both parties.

I'll send a substantial amount to signal my willingness to cooperate and trust the receiver, similar to what was shown to me.

```
{'send': 3.50}
```

reasoning:

I'm now the SENDER in Round 2. Let me think about this:

Previous round:

- I was the receiver

- The sender sent me \$3.66 (73.2% of their endowment)
- I received \$10.98
- I returned \$5.50
- My earnings from Round 1: \$5.48

Now I'm the sender:

- I receive \$5 this round
- I can send \$0-\$5
- Amount sent gets multiplied by 3x
- Receiver decides how much to return

The previous sender showed trust by sending 73.2% of their endowment. I reciprocated fairly by returning about 50% of what I received, which gave them a profit.

Now it's my turn to send. Some considerations:

1. The previous sender showed high trust (73.2%)
2. I want to maintain cooperation in this game
3. There's communication noise, so amounts may be perceived differently
4. I should consider whether to match their level of trust or adjust based on my experience

Given that:

- The previous sender trusted me with 73.2%
- I returned enough to make it worthwhile for them
- Cooperation seemed to work well

I should probably send a similar high amount to maintain cooperation. Sending around 70-75% would be:

- 70% = \$3.50
- 73.2% = \$3.66
- 75% = \$3.75

I'll send \$3.50, which is 70% - a high level of trust showing I'm willing to cooperate, slightly more conservative than the exact amount sent to me last round.

Actually, let me reconsider. The previous sender sent \$3.66 (73.2%). I could:

1. Match their trust exactly by sending the same percentage
2. Send slightly less to be conservative

...

usage:

```
{
  "input_tokens": 630,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.370000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is a disappointing outcome after I sent \$3.50 in the previous round (Round 2). It seems the sender in this round chose not to trust me at all, or perhaps they were being strategic. However, since I have \$0 to work with, my only option is:

So the receiver returned just \$0.39 from the \$10.50 they received-barely 3.7% back. In Round 3, the sender sent nothing, so there was nothing for me to return. Now moving into Round 4, I'm sending again, and I'm noticing a stark contrast: when I sent money in Round 2, I got almost nothing back, but when someone sent to me in Round 1, I returned half of what I received.

The dynamic seems to be breaking down. My attempt to reciprocate trust in Round 2 was met with minimal reciprocation, and now the sender in Round 3 didn't cooperate at all-possibly reacting to my earlier behavior or their own negative experience. For Round 4, I need to decide whether to continue trying to build cooperation or protect myself from further exploitation. Sending nothing feels safe but might perpetuate the cycle, while sending something could either restart mutual trust or leave me vulnerable again...

usage:

```
{
  "input_tokens": 894,
  "output_tokens": 1201,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.370000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent \$0, so I received \$0 after the 3x multiplication. Since I received nothing, I have nothing to return. This is straightforward.

The pattern I'm seeing:

- Round 1: I was receiver, sender sent \$3.66, I returned \$5.50
- Round 2: I was sender, sent \$3.50 (presumably got little back based on my earnings only going up slightly)
- Round 3: I was receiver, sender sent \$0, I returned \$0
- Round 4: I was sender, sent \$2.00 (presumably got little back based on my earnings staying relatively flat)
- Round 5: I was receiver, sender sent \$0, I return \$0

There seems to be a breakdown in cooperation. I can only return what I received, which is \$0.

usage:

```
{
  "input_tokens": 1094,
  "output_tokens": 234,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.370000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the recent pattern:

- Round 1: Good cooperation when someone trusted me
- Round 2: I sent \$3.50, got minimal return - lost money
- Round 3: Received \$0
- Round 4: I sent \$2.00, appeared to get nothing back - lost money again
- Round 5: Received \$0

As a sender, I've been burned twice trying to cooperate. Others are now sending \$0, suggesting trust has broken down.

However, completely abandoning all cooperation means no mutual gains are possible. I'll send a minimal amount to test if cooperation can be restored, without risking significant loss.

```
{'send': 1.00}
```

reasoning:

Looking at the pattern:

Round 1 (Receiver): Sender sent \$3.66, I returned \$5.50 - good cooperation

Round 2 (Sender): I sent \$3.50, but apparently got very little back (my earnings only went from \$5.48 to \$7.37, which is an increase of \$1.89, meaning I got back about \$5.39 total = \$5 + \$1.89 - \$3.50)

Round 3 (Receiver): Sender sent \$0, I returned \$0

Round 4 (Sender): I sent \$2.00, but apparently got nothing back (earnings stayed at \$7.37, so \$5 - \$2 = \$3, meaning I got \$0 back)

Round 5 (Receiver): Sender sent \$0, I returned \$0

Round 6 (Sender): Now deciding

The pattern shows:

- When I cooperated as receiver (Round 1), it worked well
- When I cooperated as sender (Rounds 2 and 4), I lost money or got minimal return
- Recent rounds show others sending \$0

The note mentions communication noise - amounts may differ from what was actually sent/returned. This could explain discrepancies.

However, the trend is clear: cooperation as a sender hasn't been rewarding recently. I've been burned twice as a sender.

I have a few options:

1. Send \$0 to protect myself (matching recent behavior from others)
2. Send a small amount to test if cooperation might return
3. Send a larger amount to try to restart cooperation

Given I've been hurt twice when trying to cooperate as sender, and others are sending \$0, I should probably be more protective. But completely giving up on cooperation means no one benefits.

I'll send a minimal amount - just \$1 - to test the waters without risking too much.

usage:

```
{
  "input_tokens": 1183,
  "output_tokens": 584,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.260000000000002.

This round, The sender sent you \$0.6, so you received \$1.7999999999999998. How much do you return to The sender? (0-1.7999999999999998)

Assistant output:

Looking at this situation:

- The sender sent a small amount (\$0.60, only 12% of their endowment)
- I received \$1.80 after multiplication
- This is much more cautious than earlier in the game, likely reflecting the breakdown in trust

The sender is at least testing cooperation again rather than sending \$0. I should reciprocate to encourage this, but proportionally to their cautious gesture.

If I return \$1.00:

- Sender gets: \$5 - \$0.60 + \$1.00 = \$5.40 (small gain)

- I get: $\$1.80 - \$1.00 = \$0.80$

This rewards their willingness to restart cooperation while being realistic about the small amount of trust shown.

{'return': 1.00}

Reasoning:

Let me analyze this situation:

- The sender sent \$0.60 (12% of their endowment)
- I received \$1.80 (after 3x multiplication)
- My total earnings across all rounds are \$15.26

This is a very small amount sent - much less than the 73.2% in Round 1 or even the amounts I sent in Rounds 2 and 4. This suggests either:

1. The sender is being very cautious after experiencing cooperation failures
2. Testing the waters with minimal risk
3. Communication noise might be affecting perceived amounts

Given:

- The sender is showing some willingness to cooperate (sent something rather than \$0)
- But they're being very cautious with only \$0.60
- I want to encourage cooperation but also be realistic about the small gesture

If I return a portion that:

- Gives the sender a small positive outcome
- Shows I'm willing to reciprocate even small gestures
- But doesn't give away too much given the minimal trust shown

Option 1: Return \$1.00

- Sender gets: $\$5 - \$0.60 + \$1.00 = \5.40 (gain of \$0.40)
- I get: $\$1.80 - \$1.00 = \$0.80$ (gain of \$0.80)

Option 2: Return \$0.80

- Sender gets: $\$5 - \$0.60 + \$0.80 = \5.20 (gain of \$0.20)
- I get: $\$1.80 - \$0.80 = \$1.00$ (gain of \$1.00)

Option 3: Return \$0.60

...

```
Usage: {
  "input_tokens": 1410,
  "output_tokens": 709,
  "reasoning_tokens": 0
}
```

=====

ROUND 8

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2.5; received \$7.5; Agent_1 (Aster) returned \$0.06; payoffs Agent_2 (Briar)=\$2.56, Agent_1 (Aster)=\$7.44; noise/visible: sent visible=\$0.04, received visible=\$0.12, returned visible=\$0

Action validation: {"returned": {"amount": 0.06, "clamped": false, "max": 7.5, "min": 0, "raw": 0.06}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 41.539999999999999,
"Agent_2": 38.46

}

Visible balances after round: {

"Agent_1": 17.2,
"Agent_2": 18.560000000000002

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:54:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.48.
How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the situation:

In Round 1, the sender showed significant trust by sending 73.2% of their endowment. I reciprocated fairly, and we both benefited from cooperation.

Now as the sender, I want to maintain this cooperative dynamic. The previous round demonstrated that mutual trust can lead to positive outcomes for both parties.

I'll send a substantial amount to signal my willingness to cooperate and trust the receiver, similar to what was shown to me.

{'send': 3.50}

reasoning:

I'm now the SENDER in Round 2. Let me think about this:

Previous round:

- I was the receiver
- The sender sent me \$3.66 (73.2% of their endowment)
- I received \$10.98
- I returned \$5.50
- My earnings from Round 1: \$5.48

Now I'm the sender:

- I receive \$5 this round
- I can send \$0-\$5
- Amount sent gets multiplied by 3x
- Receiver decides how much to return

The previous sender showed trust by sending 73.2% of their endowment. I reciprocated fairly by returning about 50% of what I received, which gave them a profit.

Now it's my turn to send. Some considerations:

1. The previous sender showed high trust (73.2%)
2. I want to maintain cooperation in this game
3. There's communication noise, so amounts may be perceived differently
4. I should consider whether to match their level of trust or adjust based on my experience

Given that:

- The previous sender trusted me with 73.2%
- I returned enough to make it worthwhile for them
- Cooperation seemed to work well

I should probably send a similar high amount to maintain cooperation. Sending around 70-75% would be:

- 70% = \$3.50
- 73.2% = \$3.66
- 75% = \$3.75

I'll send \$3.50, which is 70% - a high level of trust showing I'm willing to cooperate, slightly more conservative than the exact amount sent to me last round.

Actually, let me reconsider. The previous sender sent \$3.66 (73.2%). I could:

1. Match their trust exactly by sending the same percentage
2. Send slightly less to be conservative

...

usage:

```
{
  "input_tokens": 630,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.370000000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

{'return': 0}

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is a disappointing outcome after I sent \$3.50 in the previous round (Round 2). It seems the sender in this round chose not to trust me at all, or perhaps they were being strategic. However, since I have \$0 to work with, my only option is:

{'return': 0}

usage:

```
{
  "input_tokens": 805,
  "output_tokens": 164,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.370000000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: Someone trusted me significantly, I reciprocated fairly - cooperation worked

Round 2: I sent \$3.50 showing trust, but received minimal return - cooperation failed

Round 3: Someone sent \$0 - no cooperation attempted

The game mentions communication noise, which might explain some inconsistencies. However, I've seen both positive and negative outcomes from cooperation.

Given this mixed history, I'll take a cautious approach - sending enough to signal continued willingness to cooperate, but protecting myself somewhat after the Round 2 experience.

{'send': 2.00}

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)

- I received \$10.98

- I returned \$5.50

- My earnings: $\$10.98 - \$5.50 = \$5.48$

- Sender's earnings: $\$5 - \$3.66 + \$5.50 = \6.84

Round 2: I was SENDER

- I sent \$3.50

- The amount would be multiplied to \$10.50 for the receiver

- My total earnings going into this were \$5.48

- After sending, I would have $\$5.48 + (\$5 - \$3.50) = \$5.48 + \$1.50 = \6.98 before any return

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at what they were before
- The problem says my total earnings going into Round 3 were \$7.37

Wait, let me recalculate:

- After Round 1: \$5.48
- Round 2: I'm sender, I sent \$3.50
 - I start with \$5 this round
 - I send \$3.50
 - Receiver gets \$10.50
 - What did receiver return? The problem says my total earnings are \$7.37 going into Round 3

If my earnings after Round 1 were \$5.48, and going into Round 3 they're \$7.37:
 Round 2 earnings = \$7.37 - \$5.48 = \$1.89

In Round 2, I sent \$3.50, so I kept \$1.50 from my endowment
 If I gained \$1.89 total in Round 2, then I received back: \$1.89 - \$1.50 = \$0.39

So the receiver returned just \$0.39 from the \$10.50 they received-barely 3.7% back. In Round 3, the sender sent nothing, so there was nothing for me to return. Now moving into Round 4, I'm sending again, and I'm noticing a stark contrast: when I sent money in Round 2, I got almost nothing back, but when someone sent to me in Round 1, I returned half of what I received.

The dynamic seems to be breaking down. My attempt to reciprocate trust in Round 2 was met with minimal reciprocation, and now the sender in Round 3 didn't cooperate at all-possibly reacting to my earlier behavior or their own negative experience. For Round 4, I need to decide whether to continue trying to build cooperation or protect myself from further exploitation. Sending nothing feels safe but might perpetuate the cycle, while sending something could either restart mutual trust or leave me vulnerable again....

usage:

```
{
  "input_tokens": 894,
  "output_tokens": 1201,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.370000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

{'return': 0}

reasoning:

The sender sent \$0, so I received \$0 after the 3x multiplication. Since I received nothing, I have nothing to return. This is straightforward.

The pattern I'm seeing:

- Round 1: I was receiver, sender sent \$3.66, I returned \$5.50
- Round 2: I was sender, sent \$3.50 (presumably got little back based on my earnings only going up slightly)
- Round 3: I was receiver, sender sent \$0, I returned \$0
- Round 4: I was sender, sent \$2.00 (presumably got little back based on my earnings staying relatively flat)
- Round 5: I was receiver, sender sent \$0, I return \$0

There seems to be a breakdown in cooperation. I can only return what I received, which is \$0.

usage:

```
{
  "input_tokens": 1094,
  "output_tokens": 234,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.370000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the recent pattern:

Round 1: Good cooperation when someone trusted me

Round 2: I sent \$3.50, got minimal return - lost money

Round 3: Received \$0

Round 4: I sent \$2.00, appeared to get nothing back - lost money again

Round 5: Received \$0

As a sender, I've been burned twice trying to cooperate. Others are now sending \$0, suggesting trust has broken down.

However, completely abandoning all cooperation means no mutual gains are possible. I'll send a minimal amount to test if cooperation can be restored, without risking significant loss.

{'send': 1.00}

reasoning:

Looking at the pattern:

Round 1 (Receiver): Sender sent \$3.66, I returned \$5.50 - good cooperation

Round 2 (Sender): I sent \$3.50, but apparently got very little back (my earnings only went from \$5.48 to \$7.37, which is an increase of \$1.89, meaning I got back about \$5.39 total = \$5 + \$1.89 - \$3.50)

Round 3 (Receiver): Sender sent \$0, I returned \$0

Round 4 (Sender): I sent \$2.00, but apparently got nothing back (earnings stayed at \$7.37, so \$5 - \$2 = \$3, meaning I got \$0 back)

Round 5 (Receiver): Sender sent \$0, I returned \$0

Round 6 (Sender): Now deciding

The pattern shows:

- When I cooperated as receiver (Round 1), it worked well
- When I cooperated as sender (Rounds 2 and 4), I lost money or got minimal return
- Recent rounds show others sending \$0

The note mentions communication noise - amounts may differ from what was actually sent/returned. This could explain discrepancies.

However, the trend is clear: cooperation as a sender hasn't been rewarding recently. I've been burned twice as a sender.

I have a few options:

1. Send \$0 to protect myself (matching recent behavior from others)
2. Send a small amount to test if cooperation might return
3. Send a larger amount to try to restart cooperation

Given I've been hurt twice when trying to cooperate as sender, and others are sending \$0, I should probably be more protective. But completely giving up on cooperation means no one benefits.

I'll send a minimal amount - just \$1 - to test the waters without risking too much.

usage:

```
{
  "input_tokens": 1183,
  "output_tokens": 584,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.260000000000002.

This round, The sender sent you \$0.6, so you received \$1.7999999999999998. How much do you return to The sender? (0-1.7999999999999998)

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent a small amount (\$0.60, only 12% of their endowment)
- I received \$1.80 after multiplication
- This is much more cautious than earlier in the game, likely reflecting the breakdown in trust

The sender is at least testing cooperation again rather than sending \$0. I should reciprocate to encourage this, but proportionally to their cautious gesture.

If I return \$1.00:

- Sender gets: $\$5 - \$0.60 + \$1.00 = \5.40 (small gain)
- I get: $\$1.80 - \$1.00 = \$0.80$

This rewards their willingness to restart cooperation while being realistic about the small amount of trust shown.

{'return': 1.00}

reasoning:

Let me analyze this situation:

- The sender sent \$0.60 (12% of their endowment)
- I received \$1.80 (after 3x multiplication)
- My total earnings across all rounds are \$15.26

This is a very small amount sent - much less than the 73.2% in Round 1 or even the amounts I sent in Rounds 2 and 4. This suggests either:

1. The sender is being very cautious after experiencing cooperation failures
2. Testing the waters with minimal risk
3. Communication noise might be affecting perceived amounts

Given:

- The sender is showing some willingness to cooperate (sent something rather than \$0)
- But they're being very cautious with only \$0.60
- I want to encourage cooperation but also be realistic about the small gesture

If I return a portion that:

- Gives the sender a small positive outcome
- Shows I'm willing to reciprocate even small gestures
- But doesn't give away too much given the minimal trust shown

Option 1: Return \$1.00

- Sender gets: $\$5 - \$0.60 + \$1.00 = \5.40 (gain of \$0.40)
- I get: $\$1.80 - \$1.00 = \$0.80$ (gain of \$0.80)

Option 2: Return \$0.80

- Sender gets: $\$5 - \$0.60 + \$0.80 = \5.20 (gain of \$0.20)
- I get: $\$1.80 - \$0.80 = \$1.00$ (gain of \$1.00)

Option 3: Return \$0.60

...

usage:

```
{
  "input_tokens": 1410,
  "output_tokens": 709,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.0600000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent development:

Round 7 showed a positive sign - as receiver, I got a small amount (\$1.80) and returned \$1.00, demonstrating fair reciprocation. This seems to have worked out for both parties.

This small cooperation success after several rounds of \$0 exchanges suggests trust might be rebuilding. I should respond to this positive signal by increasing my send amount, but remain somewhat cautious given the earlier failures when I sent large amounts.

I'll send a moderate amount that's between my cautious \$1.00 and my overly optimistic \$3.50 from earlier.

{'send': 2.50}

Reasoning:

Let me analyze the situation:

Round history from my perspective:

- Round 1: I was receiver, someone trusted me, I reciprocated - good cooperation
- Round 2: I sent \$3.50, got minimal return - cooperation failed
- Round 3: I received \$0 as receiver
- Round 4: I sent \$2.00 as sender, likely got nothing
- Round 5: I received \$0 as receiver
- Round 6: I sent \$1.00 as sender
- Round 7: I was receiver, got \$1.80 (sender sent \$0.60), I returned \$1.00 - showed reciprocity
- Round 8: Now I'm sender again

My total earnings increased from \$15.26 to \$16.06, which is an increase of \$0.80. This matches what I calculated in Round 7 (I kept \$0.80 after returning \$1.00).

Round 7 showed a small positive cooperation signal - the sender sent \$0.60, I returned \$1.00, which gave them a gain. This is a sign that cooperation might be rebuilding.

However, I need to be cautious because:

1. I've been burned when I sent larger amounts (\$3.50 and \$2.00)
2. The amounts being sent are still quite small
3. The communication noise mentioned could be affecting things

But the Round 7 interaction was positive - I showed I would reciprocate fairly on a small amount. This might encourage more cooperation.

I should send a moderate amount - more than the \$1.00 I sent in Round 6 to show I'm willing to increase cooperation after the positive Round 7 signal, but not as much as the \$3.50 that failed in Round 2.

I'll send \$2.50 - it's a meaningful increase that signals trust while still being somewhat cautious.

```
Usage: {
  "input_tokens": 1393,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:54:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
```

```

    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.49.

This round, The sender sent you \$0.21, so you received \$0.63. How much do you return to the sender? (0-0.63)

Message 3 role=assistant

content:

Looking at this situation:

In Round 1, I sent \$3 and received a very generous return of \$6.49, suggesting the receiver was highly cooperative.

Now in Round 2, the sender only sent \$0.21 (which became \$0.63 for me). This is a very small amount - possibly due to the communication noise mentioned, or perhaps they're being cautious.

Given:

- The previous receiver was generous to me
- I want to signal willingness to cooperate
- But the sender showed minimal trust/investment
- I should be fair but proportional to what was sent

I'll return approximately half of what I received, which balances being cooperative while responding proportionally to the low amount sent.

```
{"return": 0.30}
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3. Based on my total earnings of \$8.49, let me figure out what happened:

- I started with \$5
- I sent \$3, so I kept \$2
- The sender became \$9 for the receiver (3×3)
- If they returned some amount R, I would have: $\$2 + R$
- My total earnings are \$8.49

So from Round 1, I must have: $\$2 + R = \8.49 , which means $R = \$6.49$

Wait, that seems very high. Let me reconsider.

Actually, my "total earnings across all rounds are \$8.49" means my cumulative total. In Round 1, I was the sender:

- Started with \$5
- Sent \$3, kept \$2
- Received \$9 (3×3) went to receiver
- Receiver returned some amount

If my earnings from Round 1 were \$8.49, then: $\$5 - \$3 + \text{returned} = \$8.49$

So: $\$2 + \text{returned} = \8.49

Returned = \$6.49

That's a very generous return from the receiver in Round 1!

Now in Round 2:

- I'm the receiver
- The sender sent me \$0.21, which became \$0.63 (0.21×3)
- I need to decide how much to return

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So the sender might have intended to send more or less than \$0.21.

Given that:

1. In Round 1, the receiver was very generous to me (returned \$6.49 on a \$9 pot)
2. The sender this round only sent \$0.21, which is very little (essentially not trusting me much)
3. There may be noise in the system

...

usage:

```
{
  "input_tokens": 538,
  "output_tokens": 773,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.82.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3 and received a very generous return (approximately \$6.16 based on my earnings)

Round 2: The sender sent only \$0.21, and I returned \$0.30 out of \$0.63 received - being fair and cooperative

Now in Round 3, I'm the sender again. The previous rounds showed:

- When I was generous, I was rewarded generously
- When someone sent me a small amount (possibly due to communication noise), I still returned fairly

I want to maintain a cooperative strategy and signal trust. Given the communication noise in the game, I'll send a solid amount to encourage reciprocity.

```
{"send": 3}
```

reasoning:

Let me analyze the situation:

Round 1 (I was SENDER):

- I sent \$3
- Based on my total earnings being \$8.49 after Round 2, I can work backwards
- I started Round 1 with \$5, sent \$3, kept \$2
- I received some amount back from the receiver
- Then in Round 2 as RECEIVER, I received \$0.63 and returned \$0.30, so I kept \$0.33
- Total earnings after Round 2: \$8.49
- So: \$2 + (amount returned in R1) + \$0.33 = \$8.49
- Amount returned in R1 = \$8.49 - \$2 - \$0.33 = \$6.16

Round 2 (I was RECEIVER):

- The sender sent \$0.21, I received \$0.63
- I returned \$0.30, I kept \$0.33

Now Round 3 (I'm SENDER again):

- My total earnings are now \$8.82
- This means I gained \$0.33 from Round 2 to before Round 3 decision ($\$8.82 - \$8.49 = \$0.33$)
- Wait, that doesn't make sense. Let me recalculate.

Actually, "total earnings across all rounds are \$8.82" before I make my decision in Round 3.

Let me think about this differently:

- Round 1: I sent \$3, kept \$2 initially, received something back
- Round 2: I received \$0.63, returned \$0.30, kept \$0.33

After Round 1, before Round 2, my earnings were $\$8.49 - \$0.33 = \$8.16$

So in Round 1: I kept $\$5 - \$3 = \$2$, plus I received back some amount

$\$2 + \text{return_amount} = \8.16

$\text{return_amount} = \$6.16$

The receiver in Round 1 was exceptionally generous-returning more than double what I sent.

In Round 2, my sender was cautious, only sending \$0.21, which I responded to by returning roughly half of what I received. Now I need to decide how much to send in Round 3, considering the pattern of trust and the possibility that communication issues may have affected the smaller transfer in Round 2....

usage:

```
{  
  "input_tokens": 751,  
  "output_tokens": 851,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.82.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing to return.

```
{"return": 0}
```

reasoning:

In Round 4, I'm the receiver and the sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

This is straightforward - I have \$0 to work with, so I can only return \$0.

```
{"return": 0}
```

usage:

```
{  
  "input_tokens": 970,  
  "output_tokens": 111,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.82.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a very generous return

Round 3: I sent \$3 and received \$0 back (based on my unchanged earnings)

Round 4: I received \$0 from the sender

In Round 3, despite sending \$3, I got no return - the receiver either defected or there was significant communication noise. Round 4's \$0 send might have been strategic caution.

Given this mixed experience, I'll be somewhat cautious but still attempt cooperation. I'll send a moderate amount to test if this receiver will reciprocate, while protecting more of my endowment.

```
{"send": 2}
```

reasoning:

Let me analyze what happened:

Round 1: I sent \$3 (as sender), kept \$2, received generous return

Round 2: I received \$0.63 as receiver, returned \$0.30, gained \$0.33

Round 3: I sent \$3 as sender, kept \$2

Round 4: I received \$0 as receiver, returned \$0, gained \$0

My total earnings are \$10.82.

Let me calculate:

- Round 1: Started with \$5, sent \$3, kept \$2. If my total was \$8.49 after round 1, that means I received \$6.49 back. So I ended round 1 with $\$2 + \$6.49 = \$8.49$
- Round 2: Started with \$8.49, received \$0.63, returned \$0.30, kept \$0.33. Total: $\$8.49 + \$0.33 = \$8.82$
- Round 3: Started with \$8.82, got \$5 new, sent \$3, kept \$2. If my total was \$10.82 after round 3, that means I received \$0 back. So I ended round 3 with $\$8.82 + \$2 + \$0 = \10.82
- Round 4: Started with \$10.82, received \$0, returned \$0. Total: \$10.82
- Round 5: Starting with \$10.82, getting \$5 new to send

So in Round 3, I sent \$3 and got \$0 back. That's concerning - the receiver defected completely.

In Round 4, the sender sent me \$0 (possibly in response to the communication noise or responding to something).

Now I'm the sender again in Round 5.

The pattern shows:

- Round 1: I cooperated (sent \$3), got very generous return (\$6.49)

...

usage:

```
{
  "input_tokens": 1042,
  "output_tokens": 728,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.13.

This round, The sender sent you \$0.39, so you received \$1.17. How much do you return to The sender? (0-1.17)

Message 11 role=assistant

content:

Looking at the situation:

Round 5: I sent \$2 and received approximately \$3.31 back - a good cooperative return

Now in Round 6: The sender sent \$0.39 (becoming \$1.17 for me) - another small amount, possibly due to communication noise

I want to maintain my cooperative approach. Similar to Round 2 when I received a small amount, I'll return approximately half, signaling that I'm trustworthy and willing to reciprocate even with smaller transactions.

```
{"return": 0.60}
```

reasoning:

Let me analyze the situation:

Round 5: I sent \$2 and apparently received about \$3.31 back (since my earnings went from \$10.82 to \$14.13, which is \$3.31 increase)

That's a good return - the receiver was cooperative!

Now in Round 6, the sender sent me \$0.39 (which became \$1.17 for me).

This is a small amount again, like in Round 2. It could be:

1. Communication noise affecting what they actually sent
2. The sender being cautious
3. Strategic play

Given that:

- When others have sent me small amounts, I've tried to be fair and cooperative
- In Round 2, I received \$0.63 and returned \$0.30 (about half)
- The last person who received my send was cooperative
- I want to maintain a cooperative stance

I should return a fair proportion. Since I received \$1.17, returning around \$0.50-\$0.60 would be proportional and fair, showing I'm willing to cooperate even with smaller amounts.

Let me return about half, similar to my Round 2 behavior.

usage:

```
{
  "input_tokens": 1260,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.700000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at recent rounds:

Round 5: I sent \$2 and received \$3.31 back - excellent cooperation from that receiver

Round 6: I received \$1.17 and returned \$0.60 - I was fair and cooperative

The pattern shows that when I send reasonable amounts, I often get good returns. Round 5's receiver was particularly trustworthy, returning more than what I originally sent.

I'll continue with a cooperative strategy and send a solid amount to encourage reciprocity from this new receiver.

```
{"send": 3}
```

reasoning:

Let me track my earnings across rounds:

Round 1: Started with \$5, sent \$3, kept \$2. Received generous return (about \$6.49 total earnings)

Round 2: I was receiver, got \$0.63, returned \$0.30, kept \$0.33 (total: \$8.82)

Round 3: I was sender, sent \$3, kept \$2. Received \$0 back (total stayed \$8.82, went to \$10.82)

Wait, that doesn't add up. Let me recalculate.

Actually, let me just focus on what I can infer:

After Round 4: \$10.82

After Round 5: \$14.13 (so I gained \$3.31, meaning I sent \$2, kept \$3, and received back \$3.31)

After Round 6: \$14.70 (so I gained \$0.57, meaning I received \$1.17, returned \$0.60, kept \$0.57)

So in Round 5, I sent \$2 and got back \$3.31 - good cooperation!

In Round 6, I received a small amount (\$0.39, tripled to \$1.17) and returned \$0.60, keeping \$0.57.

Now in Round 7, I'm the sender again. The previous receiver in Round 5 was quite cooperative, returning more than I sent. I should continue with a cooperative strategy.

I'll send \$3 again to maintain good cooperation.

usage:

```
{
  "input_tokens": 1431,
  "output_tokens": 475,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.14.

This round, The sender sent you \$0.04, so you received \$0.12. How much do you return to The sender? (0-0.12)

Assistant output:

Looking at the situation:

Round 7: I sent \$3 but received only about \$0.44 back - very poor cooperation from that receiver

Now in Round 8: The sender sent only \$0.04 (becoming \$0.12 for me) - an extremely small amount

This could be:

- Communication noise
- Strategic caution after being burned
- Testing behavior with minimal risk

Given the tiny amount involved and wanting to maintain a cooperative reputation despite the poor treatment in Round 7, I'll return a small proportional amount.

```
{"return": 0.06}
```

Reasoning:

Looking at the situation:

Round 7: I sent \$3 and received approximately \$2.44 back (since I went from \$14.70 to \$17.14, that's \$2.44 increase, and I had \$5 - \$3 = \$2 remaining, so total would be \$2 + \$2.44 = \$4.44, but my total is \$17.14...)

Let me recalculate:

- Before Round 7: \$14.70
- After Round 7 (before Round 8): \$17.14
- Increase: \$2.44

In Round 7 I was sender:

- Started with \$5
- Sent \$3
- Kept \$2
- Received some return

So: \$2 + return = \$2.44

Return = \$0.44

That's a very low return for sending \$3! The receiver was not cooperative at all.

Now in Round 8: The sender sent \$0.04 (becoming \$0.12 for me) - this is a very tiny amount.

Given:

1. The communication noise mentioned in the instructions
2. The sender sending such a small amount (possibly strategic or noise)
3. I've been generally cooperative throughout

I could:

- Return nothing since it's such a tiny amount
- Return a small proportional amount to signal cooperation

Given the tiny amount (\$0.12 total), I'll return a small but fair proportion. Returning about half seems reasonable to maintain my cooperative reputation.

\$0.06 would be half.

```
Usage: {
  "input_tokens": 1480,
  "output_tokens": 490,
  "reasoning_tokens": 0
}
```

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$3, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 44.539999999999999,
"Agent_2": 44.46

}

Visible balances after round: {

"Agent_1": 20.2,
"Agent_2": 18.560000000000002

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:54:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {


```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.82.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3 and received a very generous return (approximately \$6.16 based on my earnings)

Round 2: The sender sent only \$0.21, and I returned \$0.30 out of \$0.63 received - being fair and cooperative

Now in Round 3, I'm the sender again. The previous rounds showed:

- When I was generous, I was rewarded generously
- When someone sent me a small amount (possibly due to communication noise), I still returned fairly

I want to maintain a cooperative strategy and signal trust. Given the communication noise in the game, I'll send a solid amount to encourage reciprocity.

```
{"send": 3}
```

reasoning:

Let me analyze the situation:

Round 1 (I was SENDER):

- I sent \$3
- Based on my total earnings being \$8.49 after Round 2, I can work backwards
- I started Round 1 with \$5, sent \$3, kept \$2
- I received some amount back from the receiver
- Then in Round 2 as RECEIVER, I received \$0.63 and returned \$0.30, so I kept \$0.33
- Total earnings after Round 2: \$8.49
- So: $\$2 + (\text{amount returned in R1}) + \$0.33 = \$8.49$
- Amount returned in R1 = $\$8.49 - \$2 - \$0.33 = \6.16

Round 2 (I was RECEIVER):

- The sender sent \$0.21, I received \$0.63
- I returned \$0.30, I kept \$0.33

Now Round 3 (I'm SENDER again):

- My total earnings are now \$8.82
- This means I gained \$0.33 from Round 2 to before Round 3 decision ($\$8.82 - \$8.49 = \$0.33$)
- Wait, that doesn't make sense. Let me recalculate.

Actually, "total earnings across all rounds are \$8.82" before I make my decision in Round 3.

Let me think about this differently:

- Round 1: I sent \$3, kept \$2 initially, received something back
- Round 2: I received \$0.63, returned \$0.30, kept \$0.33

After Round 1, before Round 2, my earnings were $\$8.49 - \$0.33 = \$8.16$

So in Round 1: I kept $\$5 - \$3 = \$2$, plus I received back some amount

$\$2 + \text{return_amount} = \8.16

$\text{return_amount} = \$6.16$

The receiver in Round 1 was exceptionally generous-returning more than double what I sent.

In Round 2, my sender was cautious, only sending \$0.21, which I responded to by returning roughly half of what I received. Now I need to decide how much to send in Round 3, considering the pattern of trust and the possibility that communication issues may have affected the smaller transfer in Round 2....

usage:

```
{
  "input_tokens": 751,
  "output_tokens": 851,
  "reasoning_tokens": 0
}
```

Message 4 role=user**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.82.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender?

er? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing to return.

{"return": 0}

reasoning:

In Round 4, I'm the receiver and the sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

This is straightforward - I have \$0 to work with, so I can only return \$0.

{"return": 0}

usage:

```
{
  "input_tokens": 970,
  "output_tokens": 111,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.82.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a very generous return

Round 3: I sent \$3 and received \$0 back (based on my unchanged earnings)

Round 4: I received \$0 from the sender

In Round 3, despite sending \$3, I got no return - the receiver either defected or there was significant communication noise. Round 4's \$0 send might have been strategic caution.

Given this mixed experience, I'll be somewhat cautious but still attempt cooperation. I'll send a moderate amount to test if this receiver will reciprocate, while protecting more of my endowment.

{"send": 2}

reasoning:

Let me analyze what happened:

Round 1: I sent \$3 (as sender), kept \$2, received generous return

Round 2: I received \$0.63 as receiver, returned \$0.30, gained \$0.33

Round 3: I sent \$3 as sender, kept \$2

Round 4: I received \$0 as receiver, returned \$0, gained \$0

My total earnings are \$10.82.

Let me calculate:

- Round 1: Started with \$5, sent \$3, kept \$2. If my total was \$8.49 after round 1, that means I received \$6.49 back. So I ended round 1 with $\$2 + \$6.49 = \$8.49$
- Round 2: Started with \$8.49, received \$0.63, returned \$0.30, kept \$0.33. Total: $\$8.49 + \$0.33 = \$8.82$
- Round 3: Started with \$8.82, got \$5 new, sent \$3, kept \$2. If my total was \$10.82 after round 3, that means I received \$0 back. So I ended round 3 with $\$8.82 + \$2 + \$0 = \10.82
- Round 4: Started with \$10.82, received \$0, returned \$0. Total: \$10.82
- Round 5: Starting with \$10.82, getting \$5 new to send

So in Round 3, I sent \$3 and got \$0 back. That's concerning - the receiver defected completely.

In Round 4, the sender sent me \$0 (possibly in response to the communication noise or responding to something).

Now I'm the sender again in Round 5.

The pattern shows:

- Round 1: I cooperated (sent \$3), got very generous return (\$6.49)

...

usage:

```
{
  "input_tokens": 1042,
  "output_tokens": 728,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.13.

This round, The sender sent you \$0.39, so you received \$1.17. How much do you return to the sender? (0-1.17)

Message 9 role=assistant

content:

Looking at the situation:

Round 5: I sent \$2 and received approximately \$3.31 back - a good cooperative return

Now in Round 6: The sender sent \$0.39 (becoming \$1.17 for me) - another small amount, possibly due to communication noise

I want to maintain my cooperative approach. Similar to Round 2 when I received a small amount, I'll return approximately half, signaling that I'm trustworthy and willing to reciprocate even with smaller transactions.

```
{"return": 0.60}
```

reasoning:

Let me analyze the situation:

Round 5: I sent \$2 and apparently received about \$3.31 back (since my earnings went from \$10.82 to \$14.13, which is \$3.31 increase)

That's a good return - the receiver was cooperative!

Now in Round 6, the sender sent me \$0.39 (which became \$1.17 for me).

This is a small amount again, like in Round 2. It could be:

1. Communication noise affecting what they actually sent
2. The sender being cautious
3. Strategic play

Given that:

- When others have sent me small amounts, I've tried to be fair and cooperative
- In Round 2, I received \$0.63 and returned \$0.30 (about half)
- The last person who received my send was cooperative
- I want to maintain a cooperative stance

I should return a fair proportion. Since I received \$1.17, returning around \$0.50-\$0.60 would be proportional and fair, showing I'm willing to cooperate even with smaller amounts.

Let me return about half, similar to my Round 2 behavior.

usage:

```
{
  "input_tokens": 1260,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.700000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at recent rounds:

Round 5: I sent \$2 and received \$3.31 back - excellent cooperation from that receiver

Round 6: I received \$1.17 and returned \$0.60 - I was fair and cooperative

The pattern shows that when I send reasonable amounts, I often get good returns. Round 5's receiver was particularly trustworthy, returning more than what I originally sent.

I'll continue with a cooperative strategy and send a solid amount to encourage reciprocity from this new receiver.

{"send": 3}

reasoning:

Let me track my earnings across rounds:

Round 1: Started with \$5, sent \$3, kept \$2. Received generous return (about \$6.49 total earnings)

Round 2: I was receiver, got \$0.63, returned \$0.30, kept \$0.33 (total: \$8.82)

Round 3: I was sender, sent \$3, kept \$2. Received \$0 back (total stayed \$8.82, went to \$10.82)

Wait, that doesn't add up. Let me recalculate.

Actually, let me just focus on what I can infer:

After Round 4: \$10.82

After Round 5: \$14.13 (so I gained \$3.31, meaning I sent \$2, kept \$3, and received back \$3.31)

After Round 6: \$14.70 (so I gained \$0.57, meaning I received \$1.17, returned \$0.60, kept \$0.57)

So in Round 5, I sent \$2 and got back \$3.31 - good cooperation!

In Round 6, I received a small amount (\$0.39, tripled to \$1.17) and returned \$0.60, keeping \$0.57.

Now in Round 7, I'm the sender again. The previous receiver in Round 5 was quite cooperative, returning more than I sent. I should continue with a cooperative strategy.

I'll send \$3 again to maintain good cooperation.

usage:

```
{
  "input_tokens": 1431,
  "output_tokens": 475,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.14.

This round, The sender sent you \$0.04, so you received \$0.12. How much do you return to The sender? (0-0.12)

Message 13 role=assistant

content:

Looking at the situation:

Round 7: I sent \$3 but received only about \$0.44 back - very poor cooperation from that receiver

Now in Round 8: The sender sent only \$0.04 (becoming \$0.12 for me) - an extremely small amount

This could be:

- Communication noise
- Strategic caution after being burned
- Testing behavior with minimal risk

Given the tiny amount involved and wanting to maintain a cooperative reputation despite the poor treatment in Round 7, I'll return a small proportional amount.

```
{"return": 0.06}
```

reasoning:

Looking at the situation:

Round 7: I sent \$3 and received approximately \$2.44 back (since I went from \$14.70 to \$17.14, that's \$2.44 increase, and I had \$5 - \$3 = \$2 remaining, so total would be \$2 + \$2.44 = \$4.44, but my total is \$17.14...)

Let me recalculate:

- Before Round 7: \$14.70
- After Round 7 (before Round 8): \$17.14
- Increase: \$2.44

In Round 7 I was sender:

- Started with \$5
- Sent \$3
- Kept \$2
- Received some return

So: \$2 + return = \$2.44

Return = \$0.44

That's a very low return for sending \$3! The receiver was not cooperative at all.

Now in Round 8: The sender sent \$0.04 (becoming \$0.12 for me) - this is a very tiny amount

Given:

1. The communication noise mentioned in the instructions
2. The sender sending such a small amount (possibly strategic or noise)
3. I've been generally cooperative throughout

I could:

- Return nothing since it's such a tiny amount
- Return a small proportional amount to signal cooperation

Given the tiny amount (\$0.12 total), I'll return a small but fair proportion. Returning about half seems reasonable to maintain my cooperative reputation.

\$0.06 would be half.

usage:

```
{  
  "input_tokens": 1480,  
  "output_tokens": 490,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.2.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience as a sender:

Round 1: Sent \$3 ? Great return

Round 3: Sent \$3 ? Got nothing back

Round 5: Sent \$2 ? Good return (~\$3.31)

Round 7: Sent \$3 ? Very poor return (~\$0.44)

The results have been quite mixed - 2 good experiences and 2 bad ones. Communication noise makes it difficult to interpret whether receivers are being uncooperative or if noise is distorting the amounts.

Given this uncertainty, I'll take a moderate approach. I'll send \$2 - enough to potentially generate meaningful returns, but not risking too much given the 50% defection rate I've experienced.

```
{"send": 2}
```

Reasoning:

Let me track what has happened:

Round 1: I was SENDER, sent \$3, got generous return (~\$6.16 back)
Round 2: I was RECEIVER, received \$0.63 (from \$0.21 sent), returned \$0.30
Round 3: I was SENDER, sent \$3, got \$0 back (my earnings stayed at \$10.82)
Round 4: I was RECEIVER, received \$0 (sender sent \$0), returned \$0
Round 5: I was SENDER, sent \$2, got ~\$3.31 back (earnings went from \$10.82 to \$14.13)
Round 6: I was RECEIVER, received \$1.17 (from \$0.39 sent), returned \$0.60
Round 7: I was SENDER, sent \$3, got ~\$0.44 back (earnings went from \$14.70 to \$17.14)
Round 8: I was RECEIVER, received \$0.12 (from \$0.04 sent), returned \$0.06

Now Round 9: I'm SENDER again, earnings are \$17.20

Let me check: After Round 8, I had \$17.14. In Round 8 I kept \$0.12 - \$0.06 = \$0.06.
So I should have \$17.14 + \$0.06 = \$17.20. That checks out.

So the pattern is:

- When I send moderate to high amounts (2-3), I've had mixed results:
 - Round 1: sent \$3, great return
 - Round 3: sent \$3, got nothing back
 - Round 5: sent \$2, good return (\$3.31)
 - Round 7: sent \$3, got very little back (\$0.44)

It's quite mixed. There's clearly variation in receiver behavior, possibly exacerbated by communication noise.

...

```
Usage: {  
  "input_tokens": 1421,  
  "output_tokens": 778,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:54:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.3700000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

{'return': 0}

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is a disappointing outcome after I sent \$3.50 in the previous round (Round 2). It seems the sender in this round chose not to trust me at all, or perhaps they were being strategic. However, since I have \$0 to work with, my only option is:

{'return': 0}

usage:

```
{
  "input_tokens": 805,
  "output_tokens": 164,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.3700000000000001.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern so far:

Round 1: Someone trusted me significantly, I reciprocated fairly - cooperation worked

Round 2: I sent \$3.50 showing trust, but received minimal return - cooperation failed

Round 3: Someone sent \$0 - no cooperation attempted

The game mentions communication noise, which might explain some inconsistencies. However, I've seen both positive and negative outcomes from cooperation.

Given this mixed history, I'll take a cautious approach - sending enough to signal continued willingness to cooperate, but protecting myself somewhat after the Round 2 experience.

{'send': 2.00}

reasoning:

Let me think about what happened:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98
- I returned \$5.50
- My earnings: $\$10.98 - \$5.50 = \$5.48$
- Sender's earnings: $\$5 - \$3.66 + \$5.50 = \6.84

Round 2: I was SENDER

- I sent \$3.50
- The amount would be multiplied to \$10.50 for the receiver
- My total earnings going into this were \$5.48
- After sending, I would have $\$5.48 + (\$5 - \$3.50) = \$5.48 + \$1.50 = \6.98 before any return

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at what they were before
- The problem says my total earnings going into Round 3 were \$7.37

Wait, let me recalculate:

- After Round 1: \$5.48
- Round 2: I'm sender, I sent \$3.50
 - I start with \$5 this round
 - I send \$3.50
 - Receiver gets \$10.50
 - What did receiver return? The problem says my total earnings are \$7.37 going into Round 3

If my earnings after Round 1 were \$5.48, and going into Round 3 they're \$7.37:

Round 2 earnings = $\$7.37 - \$5.48 = \$1.89$

In Round 2, I sent \$3.50, so I kept \$1.50 from my endowment

If I gained \$1.89 total in Round 2, then I received back: $\$1.89 - \$1.50 = \$0.39$

So the receiver returned just \$0.39 from the \$10.50 they received-barely 3.7% back. In Round 3, the sender sent nothing, so there was nothing for me to return. Now moving into Round 4, I'm sending again, and I'm noticing a stark contrast: when I sent money in Round 2, I got almost nothing back, but when someone sent to me in Round 1, I returned half of what I received.

The dynamic seems to be breaking down. My attempt to reciprocate trust in Round 2 was met with minimal reciprocation, and now the sender in Round 3 didn't cooperate at all-possibly reacting to my earlier behavior or their own negative experience. For Round 4, I need to decide whether to continue trying to build cooperation or protect myself from further exploitation. Sending nothing feels safe but might perpetuate the cycle, while sending something could either restart mutual trust or leave me vulnerable again....

usage:

```
{  
  "input_tokens": 894,
```

```

        "output_tokens": 1201,
        "reasoning_tokens": 0
    }
Message 6 role=user
    content:
        Round 5

        You are the RECEIVER this round. Your total earnings across all rounds are $10.370000000000
        0001.
        This round, The sender sent you $0, so you received $0. How much do you return to The send
        er? (0-0)
Message 7 role=assistant
    content:
        Since the sender sent $0, I received $0 (after the 3x multiplication).

        With nothing received, I have nothing to return.

        {'return': 0}
    reasoning:
        The sender sent $0, so I received $0 after the 3x multiplication. Since I received nothing
        , I have nothing to return. This is straightforward.

        The pattern I'm seeing:
        - Round 1: I was receiver, sender sent $3.66, I returned $5.50
        - Round 2: I was sender, sent $3.50 (presumably got little back based on my earnings only
        going up slightly)
        - Round 3: I was receiver, sender sent $0, I returned $0
        - Round 4: I was sender, sent $2.00 (presumably got little back based on my earnings stayi
        ng relatively flat)
        - Round 5: I was receiver, sender sent $0, I return $0

        There seems to be a breakdown in cooperation. I can only return what I received, which is
        $0.
    usage:
        {
            "input_tokens": 1094,
            "output_tokens": 234,
            "reasoning_tokens": 0
        }
Message 8 role=user
    content:
        Round 6

        You are the SENDER this round. Your total earnings across all rounds are $10.37000000000000
        01.
        How much do you send to the receiver this round? (0-5)
Message 9 role=assistant
    content:
        Looking at the recent pattern:

        Round 1: Good cooperation when someone trusted me
        Round 2: I sent $3.50, got minimal return - lost money
        Round 3: Received $0
        Round 4: I sent $2.00, appeared to get nothing back - lost money again
        Round 5: Received $0

        As a sender, I've been burned twice trying to cooperate. Others are now sending $0, sugges
        ting trust has broken down.

        However, completely abandoning all cooperation means no mutual gains are possible. I'll se
        nd a minimal amount to test if cooperation can be restored, without risking significant
        loss.

        {'send': 1.00}
    reasoning:
        Looking at the pattern:

        Round 1 (Receiver): Sender sent $3.66, I returned $5.50 - good cooperation
        Round 2 (Sender): I sent $3.50, but apparently got very little back (my earnings only went
        from $5.48 to $7.37, which is an increase of $1.89, meaning I got back about $5.39 tota

```

l = \$5 + \$1.89 - \$3.50)
Round 3 (Receiver): Sender sent \$0, I returned \$0
Round 4 (Sender): I sent \$2.00, but apparently got nothing back (earnings stayed at \$7.37, so \$5 - \$2 = \$3, meaning I got \$0 back)
Round 5 (Receiver): Sender sent \$0, I returned \$0
Round 6 (Sender): Now deciding

The pattern shows:

- When I cooperated as receiver (Round 1), it worked well
- When I cooperated as sender (Rounds 2 and 4), I lost money or got minimal return
- Recent rounds show others sending \$0

The note mentions communication noise - amounts may differ from what was actually sent/returned. This could explain discrepancies.

However, the trend is clear: cooperation as a sender hasn't been rewarding recently. I've been burned twice as a sender.

I have a few options:

1. Send \$0 to protect myself (matching recent behavior from others)
2. Send a small amount to test if cooperation might return
3. Send a larger amount to try to restart cooperation

Given I've been hurt twice when trying to cooperate as sender, and others are sending \$0, I should probably be more protective. But completely giving up on cooperation means no one benefits.

I'll send a minimal amount - just \$1 - to test the waters without risking too much.

usage:

```
{
  "input_tokens": 1183,
  "output_tokens": 584,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.260000000000002.

This round, The sender sent you \$0.6, so you received \$1.7999999999999998. How much do you return to The sender? (0-1.7999999999999998)

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent a small amount (\$0.60, only 12% of their endowment)
- I received \$1.80 after multiplication
- This is much more cautious than earlier in the game, likely reflecting the breakdown in trust

The sender is at least testing cooperation again rather than sending \$0. I should reciprocate to encourage this, but proportionally to their cautious gesture.

If I return \$1.00:

- Sender gets: \$5 - \$0.60 + \$1.00 = \$5.40 (small gain)
- I get: \$1.80 - \$1.00 = \$0.80

This rewards their willingness to restart cooperation while being realistic about the small amount of trust shown.

{'return': 1.00}

reasoning:

Let me analyze this situation:

- The sender sent \$0.60 (12% of their endowment)
- I received \$1.80 (after 3x multiplication)
- My total earnings across all rounds are \$15.26

This is a very small amount sent - much less than the 73.2% in Round 1 or even the amounts

I sent in Rounds 2 and 4. This suggests either:

1. The sender is being very cautious after experiencing cooperation failures
2. Testing the waters with minimal risk
3. Communication noise might be affecting perceived amounts

Given:

- The sender is showing some willingness to cooperate (sent something rather than \$0)
- But they're being very cautious with only \$0.60
- I want to encourage cooperation but also be realistic about the small gesture

If I return a portion that:

- Gives the sender a small positive outcome
- Shows I'm willing to reciprocate even small gestures
- But doesn't give away too much given the minimal trust shown

Option 1: Return \$1.00

- Sender gets: $\$5 - \$0.60 + \$1.00 = \5.40 (gain of \$0.40)
- I get: $\$1.80 - \$1.00 = \$0.80$ (gain of \$0.80)

Option 2: Return \$0.80

- Sender gets: $\$5 - \$0.60 + \$0.80 = \5.20 (gain of \$0.20)
- I get: $\$1.80 - \$0.80 = \$1.00$ (gain of \$1.00)

Option 3: Return \$0.60

...

usage:

```
{
  "input_tokens": 1410,
  "output_tokens": 709,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.0600000000000002.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the recent development:

Round 7 showed a positive sign - as receiver, I got a small amount (\$1.80) and returned \$1.00, demonstrating fair reciprocation. This seems to have worked out for both parties.

This small cooperation success after several rounds of \$0 exchanges suggests trust might be rebuilding. I should respond to this positive signal by increasing my send amount, but remain somewhat cautious given the earlier failures when I sent large amounts.

I'll send a moderate amount that's between my cautious \$1.00 and my overly optimistic \$3.50 from earlier.

```
{'send': 2.50}
```

reasoning:

Let me analyze the situation:

Round history from my perspective:

- Round 1: I was receiver, someone trusted me, I reciprocated - good cooperation
- Round 2: I sent \$3.50, got minimal return - cooperation failed
- Round 3: I received \$0 as receiver
- Round 4: I sent \$2.00 as sender, likely got nothing
- Round 5: I received \$0 as receiver
- Round 6: I sent \$1.00 as sender
- Round 7: I was receiver, got \$1.80 (sender sent \$0.60), I returned \$1.00 - showed reciprocity
- Round 8: Now I'm sender again

My total earnings increased from \$15.26 to \$16.06, which is an increase of \$0.80. This matches what I calculated in Round 7 (I kept \$0.80 after returning \$1.00).

Round 7 showed a small positive cooperation signal - the sender sent \$0.60, I returned \$1.00, which gave them a gain. This is a sign that cooperation might be rebuilding.

However, I need to be cautious because:

1. I've been burned when I sent larger amounts (\$3.50 and \$2.00)
2. The amounts being sent are still quite small
3. The communication noise mentioned could be affecting things

But the Round 7 interaction was positive - I showed I would reciprocate fairly on a small amount. This might encourage more cooperation.

I should send a moderate amount - more than the \$1.00 I sent in Round 6 to show I'm willing to increase cooperation after the positive Round 7 signal, but not as much as the \$3.50 that failed in Round 2.

I'll send \$2.50 - it's a meaningful increase that signals trust while still being somewhat cautious.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$18.560000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent \$0, so I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

This is frustrating - in Round 8, I sent \$2.50 trying to rebuild cooperation after the positive signal in Round 7, but it seems I got nothing back (or the sender here is a different person sending \$0). The game continues to show inconsistent cooperation.

Usage: {

```
"input_tokens": 1443,
"output_tokens": 140,
"reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0; noise/visible: sent visible=\$0.59, received visible=\$1.77
Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.9},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```
"Agent_1": 44.539999999999999,
"Agent_2": 49.46
}
```

Visible balances after round: {

```
"Agent_1": 21.97,
"Agent_2": 23.560000000000002
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:54:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

Message 2 role=user

Round 4

How much do you send to the receiver this round? (0-5)

content:

Round 3: Someone sent \$0 - no cooperation attempted

Given this mixed history, I'll take a cautious approach - sending enough to signal continued willingness to cooperate, but protecting myself somewhat after the Round 2 experience.

reasoning:

- Sender's earnings: $\$5 - \$3.66 + \$5.50 = \6.84

- After sending, I would have $\$5.48 + (\$5 - \$3.50) = \$5.48 + \$1.50 = \6.98 before any return

- The problem says my total earnings going into Round 3 were \$7.37

- What did receiver return? The problem says my total earnings are \$7.37 going into Round 3

Round 2 earnings = \$7.37 - \$5.48 = \$1.89

If I gained \$1.89 total in Round 2, then I received back: $\$1.89 - \$1.50 = \$0.39$

The dynamic seems to be breaking down. My attempt to reciprocate trust in Round 2 was met

with minimal reciprocation, and now the sender in Round 3 didn't cooperate at all-possibly reacting to my earlier behavior or their own negative experience. For Round 4, I need to decide whether to continue trying to build cooperation or protect myself from further exploitation. Sending nothing feels safe but might perpetuate the cycle, while sending something could either restart mutual trust or leave me vulnerable again....

usage:

```
{
  "input_tokens": 894,
  "output_tokens": 1201,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.370000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent \$0, so I received \$0 after the 3x multiplication. Since I received nothing, I have nothing to return. This is straightforward.

The pattern I'm seeing:

- Round 1: I was receiver, sender sent \$3.66, I returned \$5.50
- Round 2: I was sender, sent \$3.50 (presumably got little back based on my earnings only going up slightly)
- Round 3: I was receiver, sender sent \$0, I returned \$0
- Round 4: I was sender, sent \$2.00 (presumably got little back based on my earnings staying relatively flat)
- Round 5: I was receiver, sender sent \$0, I return \$0

There seems to be a breakdown in cooperation. I can only return what I received, which is \$0.

usage:

```
{
  "input_tokens": 1094,
  "output_tokens": 234,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.370000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the recent pattern:

Round 1: Good cooperation when someone trusted me
Round 2: I sent \$3.50, got minimal return - lost money
Round 3: Received \$0
Round 4: I sent \$2.00, appeared to get nothing back - lost money again
Round 5: Received \$0

As a sender, I've been burned twice trying to cooperate. Others are now sending \$0, suggesting trust has broken down.

However, completely abandoning all cooperation means no mutual gains are possible. I'll send a minimal amount to test if cooperation can be restored, without risking significant loss.


```
{'send': 1.00}
```

reasoning:

Looking at the pattern:

Round 1 (Receiver): Sender sent \$3.66, I returned \$5.50 - good cooperation

Round 2 (Sender): I sent \$3.50, but apparently got very little back (my earnings only went from \$5.48 to \$7.37, which is an increase of \$1.89, meaning I got back about \$5.39 total = \$5 + \$1.89 - \$3.50)

Round 3 (Receiver): Sender sent \$0, I returned \$0

Round 4 (Sender): I sent \$2.00, but apparently got nothing back (earnings stayed at \$7.37, so \$5 - \$2 = \$3, meaning I got \$0 back)

Round 5 (Receiver): Sender sent \$0, I returned \$0

Round 6 (Sender): Now deciding

The pattern shows:

- When I cooperated as receiver (Round 1), it worked well
- When I cooperated as sender (Rounds 2 and 4), I lost money or got minimal return
- Recent rounds show others sending \$0

The note mentions communication noise - amounts may differ from what was actually sent/returned. This could explain discrepancies.

However, the trend is clear: cooperation as a sender hasn't been rewarding recently. I've been burned twice as a sender.

I have a few options:

1. Send \$0 to protect myself (matching recent behavior from others)
2. Send a small amount to test if cooperation might return
3. Send a larger amount to try to restart cooperation

Given I've been hurt twice when trying to cooperate as sender, and others are sending \$0, I should probably be more protective. But completely giving up on cooperation means no one benefits.

I'll send a minimal amount - just \$1 - to test the waters without risking too much.

usage:

```
{
  "input_tokens": 1183,
  "output_tokens": 584,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.260000000000002.

This round, The sender sent you \$0.6, so you received \$1.7999999999999998. How much do you return to The sender? (0-1.7999999999999998)

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent a small amount (\$0.60, only 12% of their endowment)
- I received \$1.80 after multiplication
- This is much more cautious than earlier in the game, likely reflecting the breakdown in trust

The sender is at least testing cooperation again rather than sending \$0. I should reciprocate to encourage this, but proportionally to their cautious gesture.

If I return \$1.00:

- Sender gets: \$5 - \$0.60 + \$1.00 = \$5.40 (small gain)
- I get: \$1.80 - \$1.00 = \$0.80

This rewards their willingness to restart cooperation while being realistic about the small amount of trust shown.

```
{'return': 1.00}
```

reasoning:

Let me analyze this situation:

- The sender sent \$0.60 (12% of their endowment)
- I received \$1.80 (after 3x multiplication)
- My total earnings across all rounds are \$15.26

This is a very small amount sent - much less than the 73.2% in Round 1 or even the amounts I sent in Rounds 2 and 4. This suggests either:

1. The sender is being very cautious after experiencing cooperation failures
2. Testing the waters with minimal risk
3. Communication noise might be affecting perceived amounts

Given:

- The sender is showing some willingness to cooperate (sent something rather than \$0)
- But they're being very cautious with only \$0.60
- I want to encourage cooperation but also be realistic about the small gesture

If I return a portion that:

- Gives the sender a small positive outcome
- Shows I'm willing to reciprocate even small gestures
- But doesn't give away too much given the minimal trust shown

Option 1: Return \$1.00

- Sender gets: $\$5 - \$0.60 + \$1.00 = \5.40 (gain of \$0.40)
- I get: $\$1.80 - \$1.00 = \$0.80$ (gain of \$0.80)

Option 2: Return \$0.80

- Sender gets: $\$5 - \$0.60 + \$0.80 = \5.20 (gain of \$0.20)
- I get: $\$1.80 - \$0.80 = \$1.00$ (gain of \$1.00)

Option 3: Return \$0.60

...

usage:

```
{
  "input_tokens": 1410,
  "output_tokens": 709,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.0600000000000002.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the recent development:

Round 7 showed a positive sign - as receiver, I got a small amount (\$1.80) and returned \$1.00, demonstrating fair reciprocation. This seems to have worked out for both parties.

This small cooperation success after several rounds of \$0 exchanges suggests trust might be rebuilding. I should respond to this positive signal by increasing my send amount, but remain somewhat cautious given the earlier failures when I sent large amounts.

I'll send a moderate amount that's between my cautious \$1.00 and my overly optimistic \$3.50 from earlier.

```
{'send': 2.50}
```

reasoning:

Let me analyze the situation:

Round history from my perspective:

- Round 1: I was receiver, someone trusted me, I reciprocated - good cooperation
- Round 2: I sent \$3.50, got minimal return - cooperation failed
- Round 3: I received \$0 as receiver
- Round 4: I sent \$2.00 as sender, likely got nothing
- Round 5: I received \$0 as receiver

- Round 6: I sent \$1.00 as sender
- Round 7: I was receiver, got \$1.80 (sender sent \$0.60), I returned \$1.00 - showed reciprocity
- Round 8: Now I'm sender again

My total earnings increased from \$15.26 to \$16.06, which is an increase of \$0.80. This matches what I calculated in Round 7 (I kept \$0.80 after returning \$1.00).

Round 7 showed a small positive cooperation signal - the sender sent \$0.60, I returned \$1.00, which gave them a gain. This is a sign that cooperation might be rebuilding.

However, I need to be cautious because:

1. I've been burned when I sent larger amounts (\$3.50 and \$2.00)
2. The amounts being sent are still quite small
3. The communication noise mentioned could be affecting things

But the Round 7 interaction was positive - I showed I would reciprocate fairly on a small amount. This might encourage more cooperation.

I should send a moderate amount - more than the \$1.00 I sent in Round 6 to show I'm willing to increase cooperation after the positive Round 7 signal, but not as much as the \$3.50 that failed in Round 2.

I'll send \$2.50 - it's a meaningful increase that signals trust while still being somewhat cautious.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$18.560000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent \$0, I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent \$0, so I received \$0 (after the 3x multiplication).

With nothing received, I have nothing to return.

This is frustrating - in Round 8, I sent \$2.50 trying to rebuild cooperation after the positive signal in Round 7, but it seems I got nothing back (or the sender here is a different person sending \$0). The game continues to show inconsistent cooperation.

usage:

```
{
  "input_tokens": 1443,
  "output_tokens": 140,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$18.560000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Analyzing the recent pattern:

Round 6: I sent \$1.00 (small test of cooperation)
Round 7: Received small amount, returned fairly - positive interaction
Round 8: I increased to \$2.50 to build on that cooperation
Round 9: Received \$0 - cooperation collapsed again

My earnings (\$18.56) are actually below what I'd have if I'd never cooperated at all (keeping \$5 per sender round). Every time I try to send meaningful amounts as sender, I lose money. The only successful cooperation was Round 7's small exchange.

The pattern is clear: substantial cooperation attempts are not being reciprocated. I'm consistently losing money when I send.

Given this reality, I need to protect my earnings.

```
{'send': 0}
```

Reasoning:

Let me review the recent rounds:

Round 6: I sent \$1.00 (as sender)
Round 7: I received \$1.80 and returned \$1.00 (as receiver) - good cooperation
Round 8: I sent \$2.50 (as sender) - trying to build on Round 7's cooperation
Round 9: I received \$0 (as receiver) - cooperation failed again

So in Round 8, I increased my send to \$2.50 hoping to build on the positive cooperation in Round 7, but apparently got nothing back (my earnings didn't increase from \$16.06 to \$18.56). Then in Round 9, I received \$0.

The pattern is very frustrating. Every time I try to cooperate more substantially as a sender, I get burned. The only time cooperation worked was Round 7 when someone sent me a small amount and I reciprocated fairly.

My total earnings are \$18.56, which is barely above the baseline of \$50 (5 rounds × \$5 = \$25 if I never sent anything and never received anything). Wait, let me recalculate:

If I just kept my \$5 each round as sender without sending:

- Round 1, 2, 3, 4, 6, 8, 10 (7 rounds as sender) = \$35

If I received nothing and returned nothing as receiver:

- Round 5, 7, 9 (3 rounds as receiver) = \$0

That would be \$35 total if I never cooperated.

I have \$18.56, which is significantly less. This means I've been losing money by trying to cooperate.

Given:

1. Round 8 I sent \$2.50 and apparently got minimal/nothing back

2. Round 9 received \$0

3. Every time I send a substantial amount, I lose money

...

Usage: {

"input_tokens": 1429,

"output_tokens": 676,

"reasoning_tokens": 0

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:55:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 1,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

```

    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$10.82.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent \$0, I received \$0. There is nothing to return.

```
{"return": 0}
```

reasoning:

In Round 4, I'm the receiver and the sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

This is straightforward - I have \$0 to work with, so I can only return \$0.

```
{"return": 0}
```

usage:

```
{  
  "input_tokens": 970,  
  "output_tokens": 111,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.82.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern so far:

Round 1: I sent \$3 and received a very generous return

Round 3: I sent \$3 and received \$0 back (based on my unchanged earnings)

Round 4: I received \$0 from the sender

In Round 3, despite sending \$3, I got no return - the receiver either defected or there was significant communication noise. Round 4's \$0 send might have been strategic caution.

Given this mixed experience, I'll be somewhat cautious but still attempt cooperation. I'll send a moderate amount to test if this receiver will reciprocate, while protecting more of my endowment.

```
{"send": 2}
```

reasoning:

Let me analyze what happened:

Round 1: I sent \$3 (as sender), kept \$2, received generous return

Round 2: I received \$0.63 as receiver, returned \$0.30, gained \$0.33

Round 3: I sent \$3 as sender, kept \$2

Round 4: I received \$0 as receiver, returned \$0, gained \$0

My total earnings are \$10.82.

Let me calculate:

- Round 1: Started with \$5, sent \$3, kept \$2. If my total was \$8.49 after round 1, that means I received \$6.49 back. So I ended round 1 with $\$2 + \$6.49 = \$8.49$
- Round 2: Started with \$8.49, received \$0.63, returned \$0.30, kept \$0.33. Total: $\$8.49 + \$0.33 = \$8.82$
- Round 3: Started with \$8.82, got \$5 new, sent \$3, kept \$2. If my total was \$10.82 after round 3, that means I received \$0 back. So I ended round 3 with $\$8.82 + \$2 + \$0 = \10.82
- Round 4: Started with \$10.82, received \$0, returned \$0. Total: \$10.82
- Round 5: Starting with \$10.82, getting \$5 new to send

So in Round 3, I sent \$3 and got \$0 back. That's concerning - the receiver defected completely.

In Round 4, the sender sent me \$0 (possibly in response to the communication noise or responding to something).

Now I'm the sender again in Round 5.

The pattern shows:

- Round 1: I cooperated (sent \$3), got very generous return (\$6.49)

...

usage:

```
{  
  "input_tokens": 1042,  
  "output_tokens": 728,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.13.

This round, The sender sent you \$0.39, so you received \$1.17. How much do you return to the sender? (0-1.17)

Message 7 role=assistant

content:

Looking at the situation:

Round 5: I sent \$2 and received approximately \$3.31 back - a good cooperative return

Now in Round 6: The sender sent \$0.39 (becoming \$1.17 for me) - another small amount, possibly due to communication noise

I want to maintain my cooperative approach. Similar to Round 2 when I received a small amount, I'll return approximately half, signaling that I'm trustworthy and willing to reciprocate even with smaller transactions.

```
{"return": 0.60}
```

reasoning:

Let me analyze the situation:

Round 5: I sent \$2 and apparently received about \$3.31 back (since my earnings went from \$10.82 to \$14.13, which is \$3.31 increase)

That's a good return - the receiver was cooperative!

Now in Round 6, the sender sent me \$0.39 (which became \$1.17 for me).

This is a small amount again, like in Round 2. It could be:

1. Communication noise affecting what they actually sent
2. The sender being cautious
3. Strategic play

Given that:

- When others have sent me small amounts, I've tried to be fair and cooperative
- In Round 2, I received \$0.63 and returned \$0.30 (about half)
- The last person who received my send was cooperative
- I want to maintain a cooperative stance

I should return a fair proportion. Since I received \$1.17, returning around \$0.50-\$0.60 would be proportional and fair, showing I'm willing to cooperate even with smaller amounts.

Let me return about half, similar to my Round 2 behavior.

usage:

```
{
  "input_tokens": 1260,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$14.700000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at recent rounds:

Round 5: I sent \$2 and received \$3.31 back - excellent cooperation from that receiver

Round 6: I received \$1.17 and returned \$0.60 - I was fair and cooperative

The pattern shows that when I send reasonable amounts, I often get good returns. Round 5's receiver was particularly trustworthy, returning more than what I originally sent.

I'll continue with a cooperative strategy and send a solid amount to encourage reciprocity

from this new receiver.

```
{"send": 3}
```

reasoning:

Let me track my earnings across rounds:

Round 1: Started with \$5, sent \$3, kept \$2. Received generous return (about \$6.49 total earnings)

Round 2: I was receiver, got \$0.63, returned \$0.30, kept \$0.33 (total: \$8.82)

Round 3: I was sender, sent \$3, kept \$2. Received \$0 back (total stayed \$8.82, went to \$10.82)

Wait, that doesn't add up. Let me recalculate.

Actually, let me just focus on what I can infer:

After Round 4: \$10.82

After Round 5: \$14.13 (so I gained \$3.31, meaning I sent \$2, kept \$3, and received back \$3.31)

After Round 6: \$14.70 (so I gained \$0.57, meaning I received \$1.17, returned \$0.60, kept \$0.57)

So in Round 5, I sent \$2 and got back \$3.31 - good cooperation!

In Round 6, I received a small amount (\$0.39, tripled to \$1.17) and returned \$0.60, keeping \$0.57.

Now in Round 7, I'm the sender again. The previous receiver in Round 5 was quite cooperative, returning more than I sent. I should continue with a cooperative strategy.

I'll send \$3 again to maintain good cooperation.

usage:

```
{
  "input_tokens": 1431,
  "output_tokens": 475,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$17.14.

This round, The sender sent you \$0.04, so you received \$0.12. How much do you return to The sender? (0-0.12)

Message 11 role=assistant

content:

Looking at the situation:

Round 7: I sent \$3 but received only about \$0.44 back - very poor cooperation from that receiver

Now in Round 8: The sender sent only \$0.04 (becoming \$0.12 for me) - an extremely small amount

This could be:

- Communication noise
- Strategic caution after being burned
- Testing behavior with minimal risk

Given the tiny amount involved and wanting to maintain a cooperative reputation despite the poor treatment in Round 7, I'll return a small proportional amount.

```
{"return": 0.06}
```

reasoning:

Looking at the situation:

Round 7: I sent \$3 and received approximately \$2.44 back (since I went from \$14.70 to \$17.14, that's \$2.44 increase, and I had \$5 - \$3 = \$2 remaining, so total would be \$2 + \$2.44 = \$4.44, but my total is \$17.14...)

Let me recalculate:

- Before Round 7: \$14.70

- After Round 7 (before Round 8): \$17.14
- Increase: \$2.44

In Round 7 I was sender:

- Started with \$5
- Sent \$3
- Kept \$2
- Received some return

So: \$2 + return = \$2.44
Return = \$0.44

That's a very low return for sending \$3! The receiver was not cooperative at all.

Now in Round 8: The sender sent \$0.04 (becoming \$0.12 for me) - this is a very tiny amount
.

Given:

1. The communication noise mentioned in the instructions
2. The sender sending such a small amount (possibly strategic or noise)
3. I've been generally cooperative throughout

I could:

- Return nothing since it's such a tiny amount
- Return a small proportional amount to signal cooperation

Given the tiny amount (\$0.12 total), I'll return a small but fair proportion. Returning about half seems reasonable to maintain my cooperative reputation.

\$0.06 would be half.

usage:

```
{
  "input_tokens": 1480,
  "output_tokens": 490,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.2.
How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my experience as a sender:

Round 1: Sent \$3 ? Great return
Round 3: Sent \$3 ? Got nothing back
Round 5: Sent \$2 ? Good return (~\$3.31)
Round 7: Sent \$3 ? Very poor return (~\$0.44)

The results have been quite mixed - 2 good experiences and 2 bad ones. Communication noise makes it difficult to interpret whether receivers are being uncooperative or if noise is distorting the amounts.

Given this uncertainty, I'll take a moderate approach. I'll send \$2 - enough to potentially generate meaningful returns, but not risking too much given the 50% defection rate I've experienced.

```
{"send": 2}
```

reasoning:

Let me track what has happened:

Round 1: I was SENDER, sent \$3, got generous return (~\$6.16 back)
Round 2: I was RECEIVER, received \$0.63 (from \$0.21 sent), returned \$0.30
Round 3: I was SENDER, sent \$3, got \$0 back (my earnings stayed at \$10.82)
Round 4: I was RECEIVER, received \$0 (sender sent \$0), returned \$0
Round 5: I was SENDER, sent \$2, got ~\$3.31 back (earnings went from \$10.82 to \$14.13)
Round 6: I was RECEIVER, received \$1.17 (from \$0.39 sent), returned \$0.60
Round 7: I was SENDER, sent \$3, got ~\$0.44 back (earnings went from \$14.70 to \$17.14)

Round 8: I was RECEIVER, received \$0.12 (from \$0.04 sent), returned \$0.06

Now Round 9: I'm SENDER again, earnings are \$17.20

Let me check: After Round 8, I had \$17.14. In Round 8 I kept $\$0.12 - \$0.06 = \$0.06$.
So I should have $\$17.14 + \$0.06 = \$17.20$. That checks out.

So the pattern is:

- When I send moderate to high amounts (2-3), I've had mixed results:
 - Round 1: sent \$3, great return
 - Round 3: sent \$3, got nothing back
 - Round 5: sent \$2, good return (\$3.31)
 - Round 7: sent \$3, got very little back (\$0.44)

It's quite mixed. There's clearly variation in receiver behavior, possibly exacerbated by communication noise.

...

usage:

```
{
  "input_tokens": 1421,
  "output_tokens": 778,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$20.2.

This round, The sender sent you \$0.59, so you received \$1.77. How much do you return to The sender? (0-1.77)

Assistant output:

Looking at the situation:

Round 9: I sent \$2 and my earnings only increased by \$3 (from \$17.2 to \$20.2), meaning I received \$0 back - yet another defection

Now in Round 10: The sender sent \$0.59 (becoming \$1.77 for me)

Despite being repeatedly defected on as a sender (3 out of 4 times now), I want to maintain my cooperative behavior as a receiver. This helps keep trust in the system and demonstrates reciprocity.

I'll return approximately half, consistent with my previous cooperative pattern.

```
{"return": 0.90}
```

Reasoning:

Looking at the situation:

Round 9: I sent \$2. My earnings went from \$17.2 to \$20.2, so I gained \$3.00. This means:

- I started with \$5
- I sent \$2
- I had \$3 remaining
- I received back something, and my total gain was \$3
- So I must have received \$0 back from the receiver
- Another defection!

Now in Round 10: The sender sent \$0.59 (becoming \$1.77 for me)

This is a relatively small amount, though larger than the \$0.04 and \$0.39 I've seen before. It could be due to:

- Communication noise
- Strategic caution
- A modest test of cooperation

As a receiver, I've been trying to maintain a cooperative reputation:

- Round 2: Received \$0.33, returned \$0.16 (about half)
- Round 4: Received \$0, returned \$0
- Round 6: Received \$1.17, returned \$0.60 (just over half)
- Round 8: Received \$0.12, returned \$0.06 (half)

I've been generally cooperative, returning proportional amounts. However, I just got burned again in Round 9 when someone kept everything after I sent \$2.

Given that I've been cooperative as a receiver but keep getting defected on as a sender, I should continue being a fair receiver to maintain the possibility of cooperation in the system. I'll return approximately half to slightly more than half.

$\$1.77 * 0.5 = \0.885

I'll return \$0.90, which is slightly generous but maintains cooperation.

```
Usage: {  
  "input_tokens": 1453,  
  "output_tokens": 538,  
  "reasoning_tokens": 0  
}
```